14

Manual Therapy of the Knee

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Objectives

- Identify the pertinent structure and biomechanics of the knee.
- Demonstrate the appropriate sequential knee examination.
- Identify plausible mobilization and manual therapy treatment techniques.
- Outline the special tests of the knee and the diagnostic accuracy of each of the tests.

PREVALENCE

Knee pain is present in approximately 20% of the adult population.[1] Jackson et al.[1] report that acute knee pain accounts for up to 1 million emergency room visits yearly and 1.9 million primary care visits. Knee fractures occur in less than 7% of emergency room visits.[2] A form of knee pain known as **patellofemoral pain syndrome** (PFPS) is common, accounting for 25.6% of knee-related problems.[3] A meniscal tear makes up approximately 9–11% of knee-related impairments[4] and is much less common than knee pain associated with osteoarthritis.

Selected knee injuries are variable in frequency and depend greatly on the activity level and age of the patient. Osteoarthritis significantly affects the functional capacity of knee joints in nearly 10% of adults age 60 and over.[5] Prevalence increases in individuals that are overweight,[6,7] have a history of an anterior cruciate ligament injury[8] and/or total **meniscectomy,**[7] and are females,[7] or have a genetic predisposition.[9] Nearly 44% of individuals with osteoarthritis report knee instability that is associated with a decreased ability to function during activities of daily living.[10]

Soft tissue–related injuries are common in patients who have experienced tibial plateau fractures.[11] Soft tissue injuries such as an anterior cruciate ligament tear are common during trauma or sports-related activities. Tears that are not surgically repaired run a greater risk of meniscus damage, disability, osteoarthritis, and future surgery over time.[12]

PFPS generally affects younger patients, particularly those that participate in athletic activities.[3] PFPS conditions may range from mild irritations to traumatic dislocations. Nearly half of patellofemoral dislocations occur without a first-time, predisposing incident.[13] Recurrent patellofemoral pain is uncommon (less than 15% recurrence rate) but can cause disability.[14]

Summary

- Knee problems are prevalent in up to 20% of the general population.
- Problems such as osteoarthritis are more common in individuals with previous knee injuries, are female, and have a genetic history.
- Conditions such as patellofemoral pain syndrome typically affect younger individuals who are active.

ANATOMY

Osseous Structures

The osseous structures of the knee include the femur, the tibia, and the patella. At the distal end of the femur lie the femoral condyles that are responsible for the articulation with the patella and the tibial plateau (Figure 14.1).[5] The femoral condyles are covered with articular cartilage, are convex in nature, and demonstrate two separate articulation points, one medial and one lateral.[15] The superior division of the medial and lateral condyles within the femoral groove is primarily flat. The groove deepens as distally toward the tibia.[5] The angle of the femoral groove has been associated with patellofemoral stability.[16]

The tibia (Figure 14.2) gives birth to the tibial plateau and contributes significantly to knee stability.[5] The medial and lateral plateaus are concave, a tendency that is enhanced by the inclusion of the medial and lateral menisci.[15] The lateral tibial plateau is larger to account for the movement of the lateral femoral condyle.

Between the two tibial plateaus is the pyramid-shaped intercondylar eminence. This eminence serves as a pivot point for the femur and stabilizes the knee from excessive extension.[5] This region also serves as an attachment site of the menisci.

The patella is a triangular-shaped sesamoid bone designed to improve the extensor mechanism of the knee (Figure 14.3). The inferior (posterior) surface of the patella has a medial and lateral facet but does exhibit three and occasionally four concave surfaces of articulation.[15] The medial facet typically demonstrates the greatest anatomical variation.[17] Generally, the medial facet is subdivided into two facets to more appropriately articulate with the condyle of the femurs.[17] The lateral facet is longer and wider than the medial facet and is concave in both longitudinal and medial–lateral directions.[17] Occasionally, a transverse ridge subdivides the lateral facet, creating a superior and inferior face for articulation.

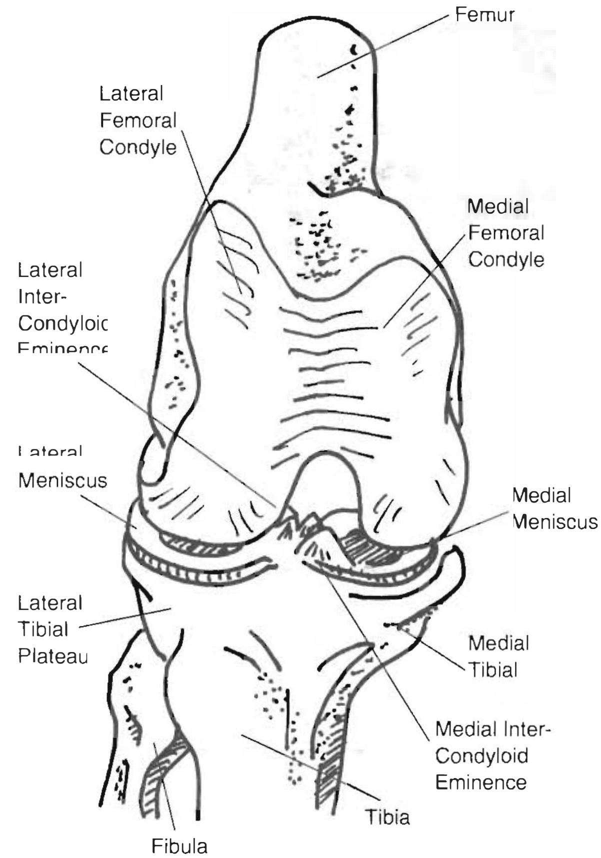

Figure 14.2 The Proximal Articulation Components of the Tibia

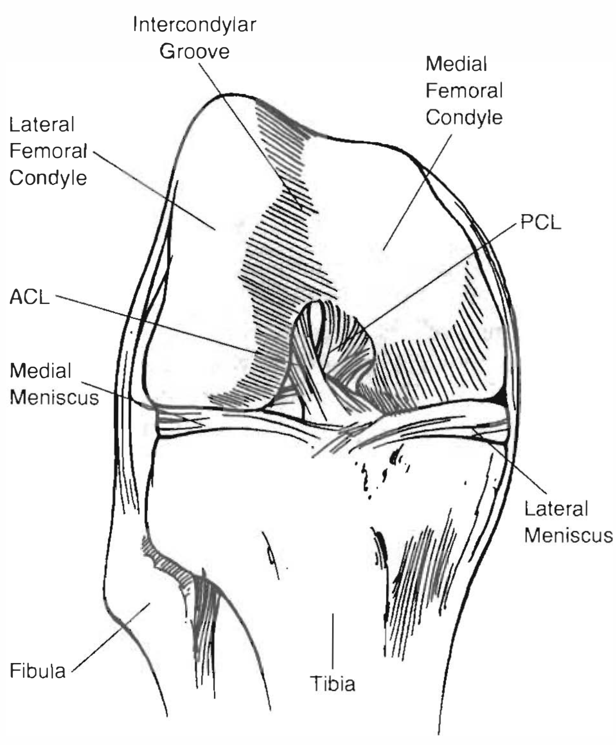

Figure 14.1 The Distal Articulation Components of the Femur

The inferior aspect of the patella articulates with the superior femoral groove during extension and the superior aspect of the patella articulates with the inferior aspect of the femoral groove during flexion.[17] The contact surface of the patella with the femur is greatest during flexion and least during full extension.[15]

Joints

The Tibiofemoral Joint

The tibiofemoral joint is the largest joint of the body (Figure 14.4). The femoral condyles are cam-shaped and circular in structure. The medial condyle of the femur is larger than the lateral condyle[15] and encounters greater weight-bearing forces than the lateral aspect.[18] Although the lateral condyle is smaller than the medial condyle, the lateral condyle projects anteriorly and acts as a stabilizer of the patella. The anterior part of the lateral condyle is flattened and provides a contact surface with the anterior horn and the anterior part of the tibial articular surface during full extension. Generally, the lateral component of the tibial

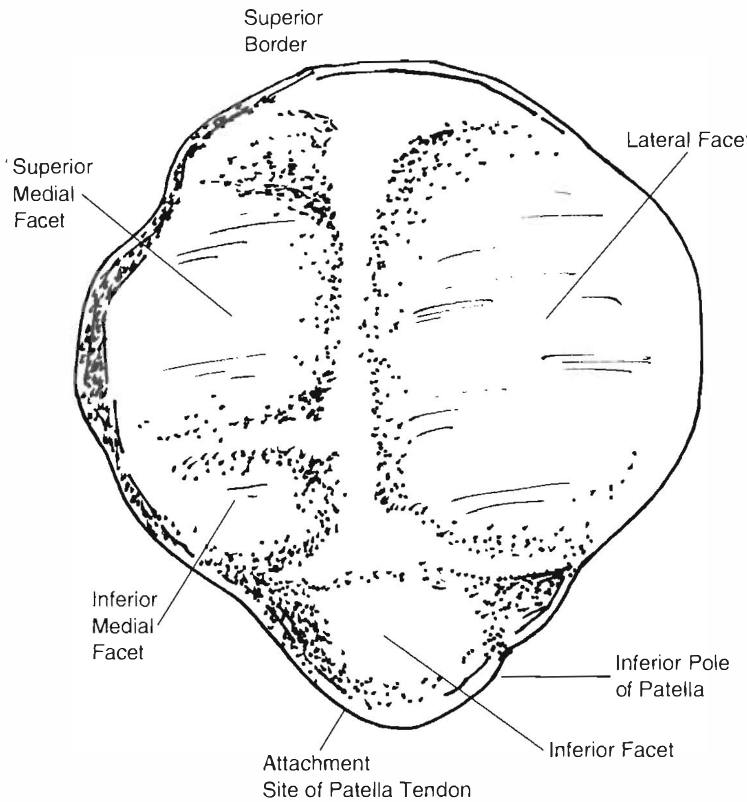

Figure 14.3 Inferior (Posterior) Surface of the Patella

Summary

- The osseous structures of the knee include the femur, the patella, and the tibia.
- The femur provides for the articulation of the patellofemoral joint and the tibiofemoral joint.
- The posterior surface of the patella promotes movement and stability within the condyle of the femur.

femoral joints allows greater mobility and is less stabile and prone to laxity more so than the medial compartment.

The femur articulates with the tibial plateau, which is inclined laterally and posteriorly and promotes stability during extension. The medial and lateral menisci, both of which contribute to stability during various degrees of extension and flexion, significantly enhance this articulation. Both menisci absorb ground reaction forces across the knee and redistribute those forces to all aspects of the femoral condyle and tibial plateau.[15]

The Patellofemoral Joint

The patellofemoral joint is the articulation of the patella within the femoral groove. During flexion and extension, the patella moves up to 7 or 8 centimeters in relation to the femoral condyles.[5] The primary function of the patella is to improve the mechanical advantage of the quadriceps during movements of flexion and extension. The patella moves in a C-shaped pattern from extension to flexion and back. Additionally, the patella tilts medially from knee flexion to laterally during knee extension. This is most likely a mechanism associated with the concurrent rotation of the tibia and the shape and congruence of the femoral condyles.

When dysfunction of the patella occurs, it may dislocate, sublux, fracture, degenerate, or develop a tracking problem.[19] Pain associated with the patellofemoral joint is typically diffuse, generally involves crepitus and locking, and may lead to decreased functional activities.[19] The tracking problem may result in a number of consequences including abnormal loading, abnormal pressures at the femur and patella, and pain.[20]

Summary

- The tibiofemoral joint is the largest joint of the body.
- The medial aspect of the femur bears more weight than the lateral aspect.
- The extremely mobile patellofemoral joint moves up to 7 or 8 centimeters in relation to the femoral condyles.

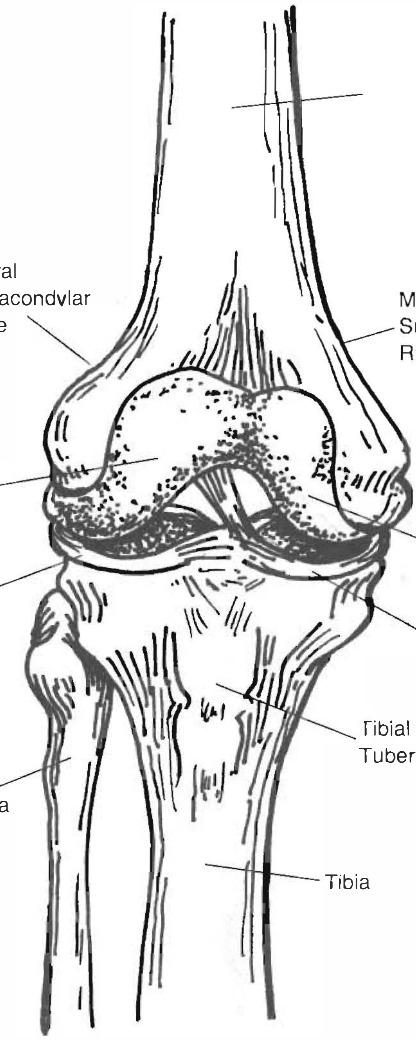

Figure 14.4 Anterior View of the Tibiofemoral Joint

Intraarticular Structures

The most prominent nonligamentous intraarticular structures are the medial and lateral menisci (Figure 14.5). The menisci's primary role is stabilization, shock absorption, proprioception, and improvement in lubrication and sequencing of the knee.[18,21] The menisci actually absorb the majority of compressive force at the knee because the combined mass is greater than that of the surrounded articular cartilage.[21] Like discs at other regions of the body, the knee menisci help to distribute contact forces over the articular surfaces by increasing the contact surface of the joint.[18]

The medial meniscus attaches anterior to the articulating surface of the tibia, to the medial aspect of the capsule of the knee and at the intercondyler tubercle. The medial meniscus also has an attachment posterior with the semimembranosus.[5] The lateral meniscus attaches (both the anterior and posterior horn) near a common attachment posterior to the intercondylar tubercle of the tibia and near the attachment of the posterior horn of the medial

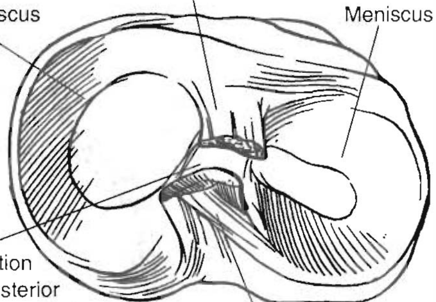

Figure 14.5 The Tibial Attachments of the Medial and Lateral Menisci

meniscus. The posterior aspect of the lateral meniscus generally is attached to two meniscofemoral ligaments and is thought to be significantly altered biomechanically by the cooperative action of the popliteus and the meniscofemoral ligaments.[22,23] Both the anterior aspects of the medial and lateral meniscus are attached via the ligamentum transversum, also known as the transverse meniscal ligament.[5]

The medial meniscus is C-shaped and the lateral meniscus is primarily circular-shaped. Both menisci distort significantly (elongate in a sagittal plane) and move posteriorly during flexion and move anteriorly during extension. This distortion is greatest during higher loads.[18] During rotation, such as during the screw home mechanism, the menisci follow the movements of the femur.[15] Actions such as weight bearing increase the mobility of the menisci. Of the two menisci, the medial meniscus encounters more forces during weight bearing than the lateral meniscus.

The menisci are damaged occasionally during weight-bearing activities and during normal processes of degeneration. Recently, studies have shown that degeneration and tears often occur synonymously.[21] Damage to the cartilage generally begins at the surface then extends through the thickness of the cartilage.[18] Meniscal damage leads to cartilage thinning and subsequent loss of joint space.[21]

The outer aspect of the menisci is innervated and is capable of producing pain when torn or degenerated.[18] Historically, damage to the menisci involved removal of the menisci to avoid the painful consequences associated with knee locking. Nonetheless, when the menisci are removed, forces along the femur and the tibia are significantly increased, specifically at the contact points of articulation.[21] Furthermore, removal of the menisci commonly leads to articular surface breakdown of the femur and tibia within a few years.[18,21,24] Articular breakdown may result in flattening of the femur at the compression site, fibrillation, sclerosis, and further narrowing of the joint space.[25]

Muscles

Several muscles contribute to normal knee motion. The quadriceps femoris muscles include the rectus femoris, vastus lateralis, vastus medialis, and the vastus intermedialis. Much of the efficiency of the knee extensor mechanism depends on the timing and position of the patella during muscular contraction. The maximal quadriceps contraction occurs at 60 degrees of knee flexion.[15]

The knee flexors include the hamstrings, sartorius, gracilis, popliteus, and the gastronemius muscles. The popliteus muscle also contributes to knee flexion concurrently while initiating internal rotation.

Summary

- The medial and lateral menisci's primary role is stabilization, shock absorption, proprioception, and improvement in lubrication and sequencing of the knee.
- Both menisci distort significantly (elongate in a sagittal plane) and move posteriorly during flexion and move anteriorly during extension.
- The outer aspect of the menisci is innervated and is capable of becoming a pain generator.
- The collective output of the muscles in the knee work in concert to increase the stability during static and dynamic activities.

Ligaments and Capsule of the Knee

The Anterior Cruciate Ligament

The anterior cruciate ligament (ACL) is the primary structure responsible to counter anterior tibial transfer with respect to the femur (Figure 14.6).[5] The ACL is a broad helical ligament that allows for appropriate tibia-to-femur rotation during movement but still supports the transfer of the tibia and side-to-side stabilization of the knee. The femoral attachment of the ACL is at the posterior aspect of the medial surface of the lateral femoral condyle.[5] The tibial attachments are at the anterior aspect of the tibial spine.[5]

Like other ligaments of the knee, the ACL is actually two distinct bands, an anterior-medial and a posterolateral band, based on the origins of the tibia. The anterior-medial band is taut in flexion and the posterolateral band is taut in extension.[5] Subsequently, the bands change tension throughout movement of the knee from flexion to extension. Since the ligaments are relatively horizontal during flexion, the ligaments serve to stabilize tibial translation mostly at nearly 90 degrees. Nonetheless, the ligament is also a secondary restraint to varus and valgus forces and internal rotation of the tibia.

The ligament is commonly injured when the tibia is driven anteriorly on the femur but a large portion of ACL injuries are nontraumatic, occurring commonly in female athletes

the ACL may lead to increased risk of meniscus, chondral, and subchondral injury within 5 to 10 years.[12]

The Posterior Cruciate Ligament

The posterior cruciate ligament (PCL) has a relatively compact tibial attachment, located posteriorly between the posterior horn of the medial and lateral menisci.[27] The femoral attachment is extensive adjacent to the condylar cartilage and near the medial femoral condyle.[27] Most researchers agree that the PCL is made of two separate systems of fibers that function separately based on the forces provided to the knee joint.[28] The PCL fibers run anterior and proximal and are designed to counter tibial posterior translation. It is projected that the PCL is strongest while one is younger, slowly weakening with age.[27]

The PCL is the primary restraint of the tibia on the femur when the knee is flexed from 30 to 90 degrees.[27] Isolated cutting of the PCL minimally affects posterior tibial translation while the knee is in extension.[29] Structures other than the PCL are responsible for stability of

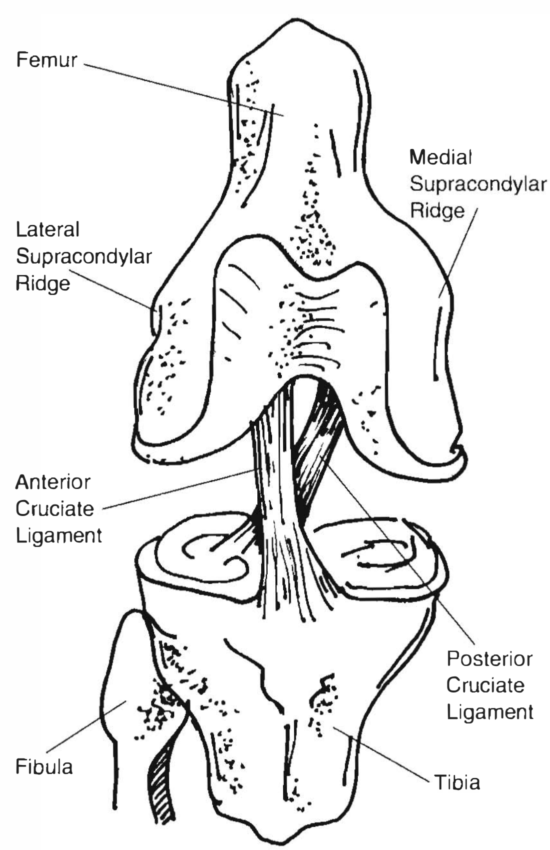

Figure 14.6 The Anterior and Posterior Cruciate Ligaments

the knee from 20 degrees of flexion to extension.[27] The PCL is a secondary restraint to tibial internal and external rotation and valgus and varus forces.[27]

The condition of the ACL affects the PCL as does the condition of the PCL affect the ACL. A stiffer ligament leads to increased force across both ligaments and a more flexible or lax ligament leads to diminished forces along both ligaments.[30] The ligaments function in concert to control anterior and posterior tibial translation.

The Lateral Collateral Ligament

The lateral collateral ligament is tightened during extension and is slackened during approximately 30 degrees of knee flexion (Figure 14.7).[29] The popliteofibular ligament (not pictured) contributes to posterior–lateral stability during all angles of flexion and contributes as a deterrent to varus forces during extension and flexion.[29]

The Medial Collateral Ligament Complex

The medial collateral ligament complex (MCL) (Figure 14.8) consists primarily of three main structures: the longitudinal fibers of the superficial medial collateral ligament (sMCL), the deep medial collateral ligament (dMCL), and the posteromedial capsule (PMC).[31] At present, the true function of these fibers and whether the fibers function together or independently is unknown.

The MCL structures provide coronal shear stability and reduce the forces associated with valgus. In additional to the passive structures, the contribution of the muscles of the pes anserine aid in stabilizing the knee. Passively, the posteromedial capsule provides passive support during extension of the knee and provides some stability at extension. The MCL contributes more toward valgus stability of the knee during flexion.

The Meniscofemoral Ligaments

The **meniscofemoral ligaments** connect to the posterior horn of the lateral meniscus to the intercondylar area of the femur.[32] Less common are meniscofemoral ligaments that attach to the anterior horn of the medial meniscus. These ligaments are present in approximately 90% of individuals. MFLs are considered stabilizers of the knee, specifically stabilizers in rotation. Others[33] indicate that MFLs are responsible for moving the posterior horn

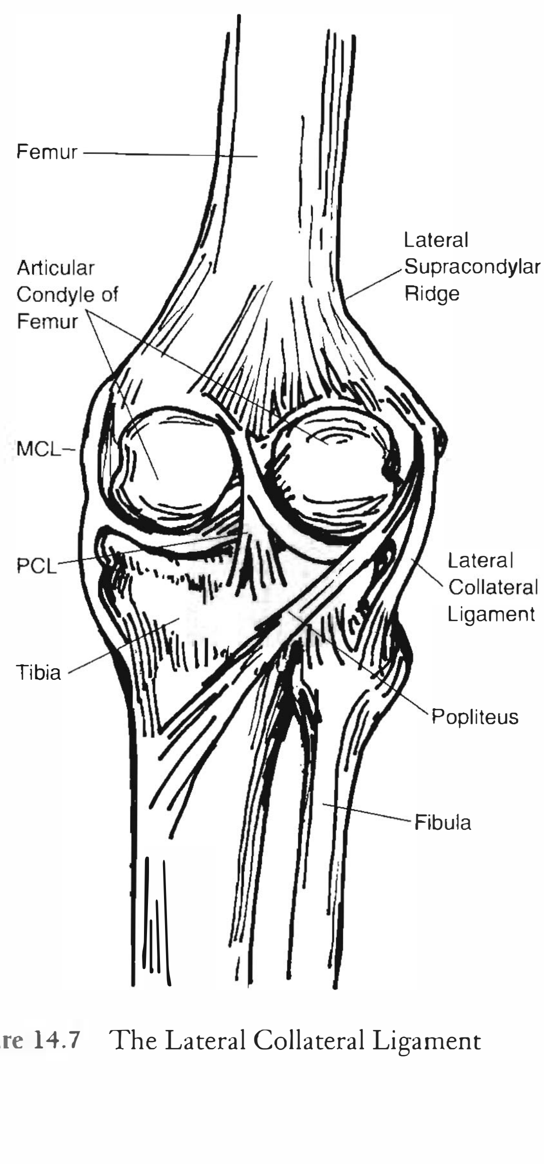

Figure 14.7 The Lateral Collateral Ligament

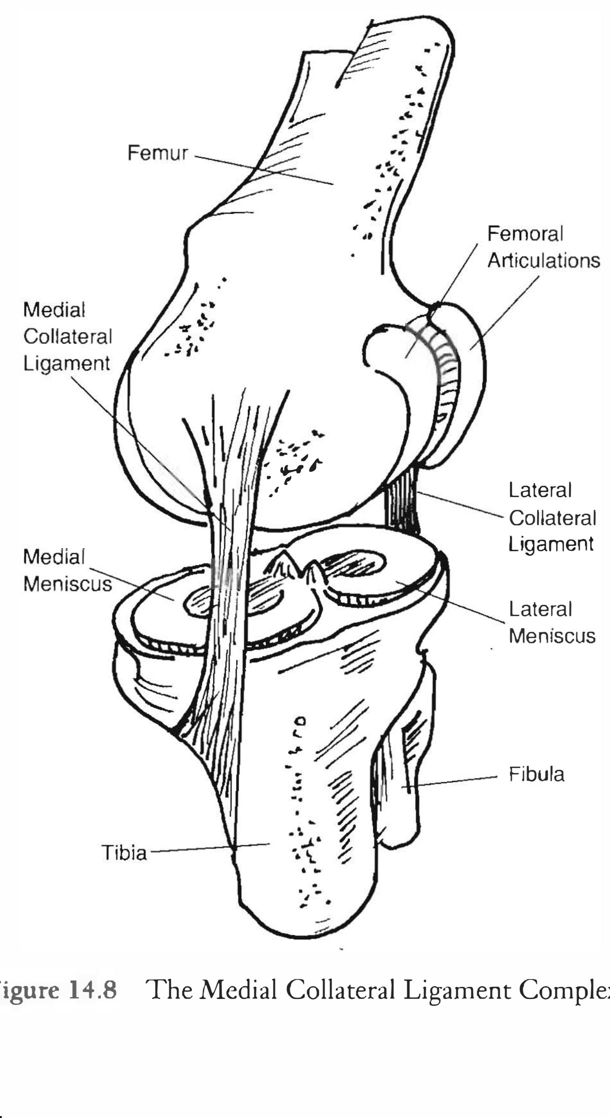

Figure 14.8 The Medial Collateral Ligament Complex

of the meniscus during movement of the knee. The ligaments keep the meniscus moving consistently with the movement of the femur, thus preventing the meniscus from being "ground" between the femur and the tibia.

Since the popliteus muscle also contributes to movement of the posterior horn of the meniscus, some have proposed that the MFLs function to counter the pull of the popliteus muscle.[34] The anterior and posterior MFLs may serve to move the meniscus during extension and flexion movement, each being responsible for the movement of the menisci depending on the position of the femur relative to the tibia. Additionally, the MFLs are considered secondary restraints to a posterior drawer of the knee.

The Posterolateral Knee Structures

The posterolateral knee structures include the lateral collateral ligament, the arcuate ligament, the popliteofibular ligament, and variably the fibellofibular ligament.[35] Isolated injuries to the posterolateral corner are rare,[35] but may occur during contact to the anterior-medial aspect of the extended knee during weight bearing[27] or forced external rotation during a loaded and extended knee.[36] When injured, patients report a sensation of knee instability during extension. Movements such as hyperextension are vague clinical findings during examination and may not be definitive of posterolateral knee instability.[35] Posterolateral knee instability is rotational in nature and may involve displacement of the tibia posteriorly on the femur.

Because the PCL couples with external rotation of the tibia during posterior translation, the PCL ligament contributes poorly to the stability of the posterior lateral corner, specifically in extension. Isolated rupture of the posterior–lateral corner has a significant effect on the rotational and translatory stability of an extended knee.[27] Because of this, posterior drawer tests performed in flexion are not sufficient to determine posterior–lateral instability. Generally, to determine posterior–lateral corner instability, the clinician must examine tibial external rotation during 30–90 degrees of knee flexion.[27,37]

Posterolateral instability may involve an injury to the popliteus tendon since this structure is considered an important restraint to tibial external rotation.[37] The popliteus tendon originates on the posterolateral aspect of the femur and inserts on the anterior–lateral aspect of the fibular head, just distal to the lateral joint line. The tendon tightens during extension and external rotation.

Posterior Ligaments and Structures

The most important posterior stabilizing structures of the knee are the deep posterior capsule, the lateral condylar ligament, the oblique popliteal ligament, the semimembranosus muscle, and the arcuate ligament. The deep posterior capsule is tightened during knee extension and blends nicely with the tendinous insertions of the gastronemius muscles. The lateral condylar ligament, oblique popliteal ligament, and the arcuate ligament assist in stabilizing the structure of the posterolateral knee. The semimembranosus

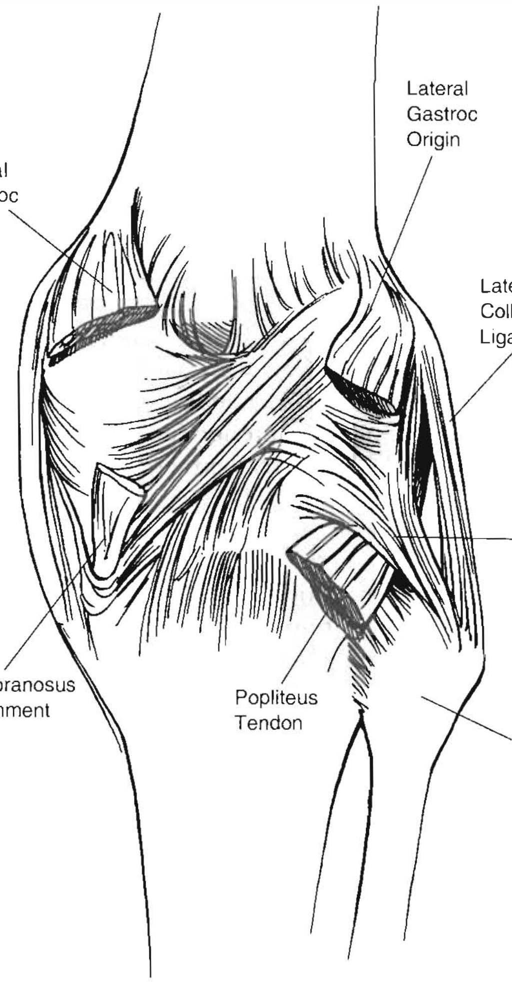

Figure 14.9 The Posterior Structures of the Knee

> **Summary**
> - There are multiple ligamentous structures responsible for stability of the knee.
> - Damage to one of the ligaments may negatively affect other ligaments.
> - The most common outcome to damaged ligaments is instability.

can contract and tighten the oblique popliteal ligament that is considered a continuation of the tendon (Figure 14.9).

Biomechanics

Movement and Stability

Tibial Femoral Joint

In a normal knee, approximately 0 to 140 degrees of extension to flexion are present.[15] Most individuals also exhibit some degree of knee hyperextension during passive weight bearing and non–weight bearing. In normal individuals, the combined contribution of the muscles, ligaments, and menisci serve to increase the stiffness and stability of the knee and allow the action of transmission of large loads throughout the joint.[30]

Tibiofemoral translation is somewhat predictable during movements of extension to flexion. As a whole, during knee flexion, the tibia moves posteriorly on a fixed femur and the menisci move posteriorly as well. During knee extension, the tibia moves anteriorly on the fixed femur and the menisci move anteriorly. Movements in weight bearing initiate structures such as the cruciate ligaments and the menisci to balance the shear and compression forces at the tibial femoral joint.[38] Anterior forces are restrained primarily by the ACL whereas posterior forces are restrained by the PCL.

Von Eisenhart-Rothe et al.[39] reported that the femur translates posteriorly on the tibia at 30–90 degrees of knee flexion during co-contraction of the extensors and flexors in a simulated loading response. However, isolated contraction on healthy subjects demonstrates anterior translation of the tibia with respect to the femur during extensor contraction and posterior translation of the tibia with respect to the femur during flexor contraction.[40] Individuals with a damaged ACL demonstrated posterior translation during isolated extensor and flexor contractions.[39] These findings are supported by other studies[41] that also found posterior migration of the femur on the tibia during flexion.

Although the lateral condyle is often considered to translate more so than the medial condyle, research results are mixed.[39,40] Rolling is a fundamental movement of most articular surfaces and the femur does roll significantly on the fixed tibia. During flexion, the femur rolls and slides posteriorly to the tibia to allow the distal and posterior aspects of the femoral condyle to interface with the flattened tibial plateau. Much of this movement is guided by the posterior translation of the menisci and the control supplied by the cruciate ligaments and can be disrupted significantly during damage to these structures.[39] The ACL pulls on the femur and encourages anterior slide during knee flexion while it rolls in a posterior direction. The PCL encourages the femoral condyle to slide posterior during active extension and roll in an anterior direction.

During extension and flexion, the knee exhibits coupled knee internal and external rotation and sagittal motions.[30] This concurrent action of coupled tibial rotation, called the **screw home mechanism**, occurs to enhance the stability of the knee. When the knee is flexed in a closed chain, the tibia moves internally, in some cases up to 6.5–36 degrees.[42,43] During knee extension in a closed chain, the tibia moves into external rotation. Conversely, the femur appears to translate posteriorly, rotate laterally, and migrate proximally during flexion with respect to the tibia.[30] To some extent, coupled varus and valgus motion occurs in conjunction with the sagittal and transverse plane movements. Nonetheless, these motions do not appear to be as compelling as rotations, flexion, and extension.[30]

During flexion beyond 120 degrees, the femoral condyles lose congruency with the tibial plateau and have contact points at the posterior horn of the menisci.[38] This is because the rolling of the femur outdistances the concurrent slide of the tibiofemoral complex. The external rotation that occurs at the femur increases the surface area for roll since it mimics the translatory aspect at the joint. Martelli and Pinskerova[44] advocate that the medial condyle is congruent and rounded and the lateral surface is flattened, allowing the lateral condyle to roll and slide at a greater quantity than the medial surface.

Screw Home Mechanism

The screw home mechanism appears to be guided by the location of the joint axis and by the passive action of the posterior and anterior cruciate ligaments.[45] This mechanism is altered when damage is incurred to the anterior and posterior cruciate ligaments or when the stress–tension relationship of this ligament has in some way been altered.[45]

The popliteus plays a vital role in the screw home mechanism and initiates rotation of the tibia at 0–20 degrees of flexion. During this rotation, the menisci follow the movements of the femur.[15] The popliteus is situated and well adapted to prevent tibial external rotation during knee flexion.[30] During active extension, the popliteus functions actively to initiate knee flexion and retraction of the lateral meniscus.[30]

Summary

- During an extension contraction, the tibia translates anteriorly.
- During a flexion contraction, the tibia translates posteriorly.
- During flexion, the tibia follows the convex–concave rule and rolls posteriorly. During extension, the tibia rolls anteriorly.
- The popliteus plays a vital role in the screw home mechanism and initiates rotation of the tibia at 0–20 degrees of flexion.
- During screw home rotation, the menisci follow the movements of the femur.

Patella Femoral Joint

Movement of the patellofemoral joint is a complex kinematic process. Numerous *in vitro* studies have investigated patellofemoral movements and have reported significant variations of patterns.[46,47] One study attempted to load

Figure 14.10 Movement of the Patella during Extension to Flexion of the Knee

the knee during 90 degrees of flexion to extension and found consistency of the patella during movement on the femur. During movement from flexion to extension, the patella starts in a medially tilted position, and then shifts to neutral, lastly shifting to a laterally tilted position while in extension (Figure 14.10).[48]

Recently, an *in vivo* study demonstrated variability in patellofemoral movements during extension to flexion movements. Laprade and Lee[3] reported that the patella typically moves distally during progressive extension to flexion, variably moves from anterior to posterior movement (in many cases the patella just moved posteriorly), and variably moves progressively laterally during knee flexion. In some cases, the kneecap moved medially first, then laterally as knee flexion progressed.

Summary

- The articulation pattern of the patella may be variable during flexion and extension movements.
- The patella typically moves distally during progressive extension to flexion, variably moves from anterior to posterior movement (in many cases the patella just moved posteriorly), and variably moves progressively laterally during knee flexion.
- Abnormalities in patellofemoral biomechanics generally manifest at 0–30 degrees of flexion because of the deepening of the trochlear groove of the femur.

Abnormalities in patellofemoral biomechanics generally manifest at 0–30 degrees of flexion because of the deepening of the trochlear groove of the femur. Increases in knee flexion increase the weight-bearing aspect of the patellofemoral joint.[49] As the tibia continues to flexion, the iliotibial band pulls posteriorly on the patella, promoting posterior displacement against the femur.[17] If the patella lies too far laterally during this motion, this process can cause abnormal lateral forces, tracking problems, and lateral dislocations. Powers[16] reported that the depth of the trochlear groove is highly correlated with abnormal patellar kinematics, with increased shallowness of the trochlear groove being predictive of lateral patellar tilt and abnormal tracking.

Assessment and Diagnosis

Subjective Considerations

The primary approach to the subjective exam should first include the possible determination of nonmechanical disorders of the knee and isolation of the type of mechanical disorder if present. With knee disorders, identifying the mechanism of injury and/or onset of pain can aid in determining if the presenting disorder may have involved possible tearing or rupturing of structures or, if the onset of pain is more insidious, a more degenerative or sinister condition may be present. Additionally, the subjective exam will serve to identify potential movements or activities that are related to the concordant signs. For example, patients with meniscal injuries often report knee pain after twisting their leg while bearing full weight and often will experience a popping or tearing sensation, followed by severe pain. Swelling can take several hours to appear. Ligamentous injuries may occur with a similar mechanism to meniscal tears, and often will be the result of a direct blow to the knee, but swelling often occurs immediately.[1] Manske and Davies[50] noted the subjective variables of pain in the region of the patellofemoral joint while ascending or descending stairs may be indicative of PFPS.

Based on criteria of the *American College of Rheumatology*, osteoarthritis of the knee may be suspected if the following criteria are present: age older than 50, knee stiffness for less than 30 minutes, crepitus, bony tenderness, bony enlargement, and no palpable warmth.[1,51] If four criteria are present, the sensitivity of the characteristics is 84% with a specificity of 89%. If at least three of the criteria are present, the chance of osteoarthritis being present is 62%. Of the five criteria, two of the criteria are considered aspects of a subjective examination (age and stiffness) and two of five criteria increase the probability of osteoarthritis to 4%.[1] With this said, at least one physical exam variable would need to be present to bring probability to acceptable levels. This is in line with the fact that the clinical history alone, related to the diagnosis of meniscus or ligamentous tears, can only heighten suspicion and help formulate tentative management strategies and not necessarily differentiate between the two. The physical exam will be more helpful in determining the pathology associated with these disorders.[1]

Nonmechanical disorders such as septic arthritis will present as an acute onset of knee pain with effusion and warmth; signs and symptoms of infection will be present as well. There will most likely be no history of trauma to correlate to the effusion. Patients with this presentation should immediately be referred to a medical physician for appropriate consultation.[1,52]

Functional scales are designed to measure the outcomes of an intervention. The *Lower Extremity Function Scale* (LEFS) is a region-specific scale that is not limited to use with osteoarthritis. The LEFS is a self-report mechanism, primarily designed as a performance-based measure.[53] The LEFS is reliable (ICC = 0.95),[54] demonstrates construct validity, and is more sensitive to change than the SF-36.[55] The LEFS is a functional scale that can measure multiple joints[54] and is used conjunctively with measures of pain and physical exertion.

Summary

- The purpose of the subjective examination is to identify the concordant mechanical disorders associated with the knee.
- Based on criteria of the *American College of Rheumatology*, osteoarthritis of the knee may be suspected if the following criteria are present: age older than 50, knee stiffness for less than 30 minutes, crepitus, bony tenderness, bony enlargement, and no palpable warmth.
- Nonmechanical disorders often include acute onset of knee pain with effusion, warmth, and signs and symptoms of infection.
- The *Lower Extremity Functional Scale* is a valid, self-report, region-specific questionnaire that measures problems associated with activities of daily living.

Clinical Examination

Observation

Observation of gait parameters and static alignment of the lower extremities are often key components to the clinical examination of the knee. The current literature regarding static alignment focuses on alignment at the foot and ankle. Selfe[56] reviewed the effect of correction of foot and ankle asymmetries and found that patients experienced an average of 67% reduction in patellofemoral knee pain following the dispensation of foot orthotics. Additionally, rear foot posting produced an immediate and statistically significant medial glide in the patellofemoral joint. Selfe[56] noted that while the rear foot posting produced the medial glide, there was not an analysis of the association to reduction in symptoms.

Gross et al.[57] reviewed the effect of foot orthoses on the patellofemoral joint and their findings indicate that a significant amount of patients report improvement in their pain; however, there is less clear evidence of how correction of foot and ankle symmetry affects the patellofemoral joint. Hinterwimmer[58] found no difference in patellar kinematics in individuals with lower extremity asymmetries of genu varum and mild medial compartment osteoarthritis. Livingston et al.[59] found that the magnitude of right to left rear foot asymmetries was no different between symptomatic and asymptomatic patellofemoral pain groups.

Observational analysis of gait deviations is another component to assessment of the patient with knee pain. In the absence of obvious deviations that occur during significant trauma and swelling, minor gait deviations often are concurrent with knee pain. Selfe[56] reviewed gait analysis in patients with patellofemoral pain and found that men and women have similar torque generation at the knee during gait, wider-soled shoes increased knee flexor torque by 30%, and patients have a decreased walking velocity and more extended knees during gait than healthy controls. Additional studies[60,61] have demonstrated similar findings. A limitation of many of these studies is that they are laboratory based or utilize special equipment to measure gait deviations. This technology is not readily available in clinical practice, thus gait deviations may need to be observable and subjectively related to the patient's complaint.

Summary

- Posture and foot alignment may suggest knee pain associated with other factors; however, in isolation, asymmetry does not implicate the causal problem.
- A gait analysis is helpful to outcome PFPS and other related knee anomalies.

Active Physiological Movements

Active movements during a clinical examination are utilized to identify physical impairments that are relevant to the concordant signs. By determining the behavior of the concordant signs to selected active movements, the clinician can effectively identify potential active physiological treatment approaches.

Plane-Based Active Range of Motion

Assessment of active range of motion of the knee can be assessed in different manners that will help not only determining objective range of motion of the knee, but also the

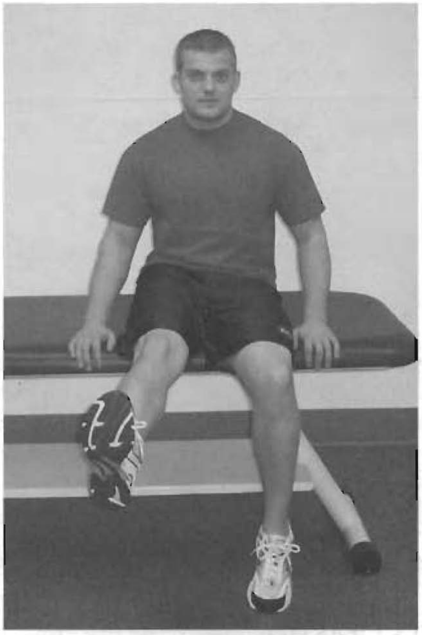

Figure 14.11 Active Physiological Knee Extension in Sitting

relativity of the impairment to the concordant signs. Non-weight bearing active range-of-motion movements (AROM) can be performed in sitting (Figures 14.11 and 14.12) or supine position. These positions will influence the stressing of tissues surrounding the knee in different ways.

Functional Active Range-of-Motion Testing

AROM performed in a weight-bearing position, while less convenient to attain objective measurements of ROM, allows for a closer replication of functional positions and movements that are relevant to the concordant signs. For example, pain-reproducing movements for the knee may often include ascending and descending stairs and squatting. Cliborne et al.[62] used a *functional squat test* as one outcome measure following a program of hip mobilization in patients with knee osteoarthritis. The procedure included having the patient stand with their feet comfortably apart, then squat down until either pain limits the range or the heels come off the floor (Figure 14.13).

A single step-down test (Figure 14.14) is used to measure functional control of the knee during eccentric and concentric knee movements. The single step-down test is useful for patients who lack the range of motion to perform the functional squat test.

Hop tests (Figure 14.15) are often used with patients that are post-ACL reconstruction as a means to determine functional activity tolerance and predict dynamic knee stability and possible future injury. The *Hop tests* appear to have potential to predict dynamic knee stability, but currently the literature suggests that the predictive capability of these tests is questioned.[63]

Summary

- Active range-of-motion techniques should include functional activities and plane-based movements.
- Functional tests should be geared toward the concordant reproduction of the patient's complaint.

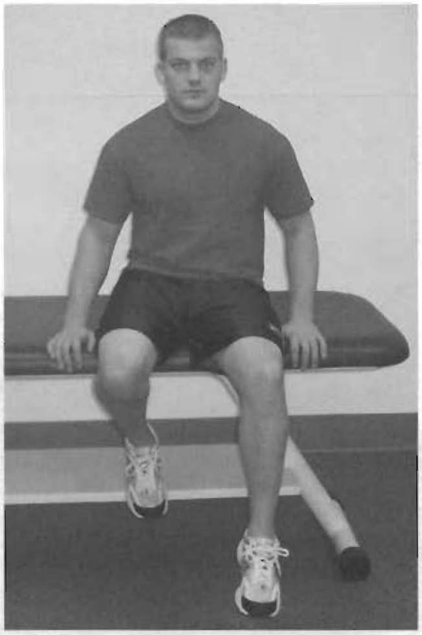

Figure 14.12 Active Physiological Knee Flexion in Sitting

Figure 14.13 The Functional Squat Test

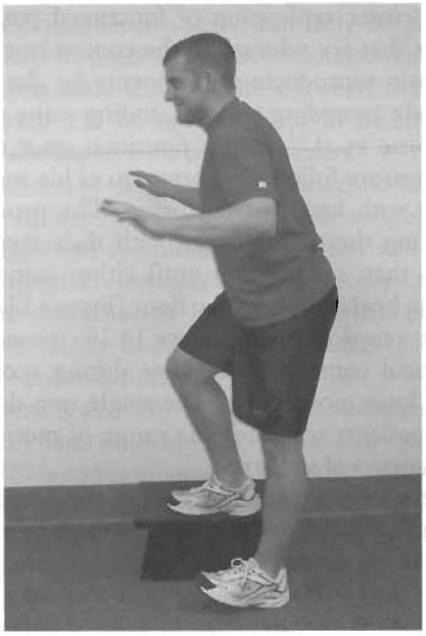

Figure 14.14 The Single Step Down Test

Figure 14.15 The Hop Test

Passive Movements

Passive Physiological Movements

Passive physiological movements of the knee are similar to active physiological movements and are used to confirm the relationship to the concordant signs. Additionally, passive physiological movements of the knee allow for a more complete examination of component movements (i.e., tibial rotation) that are difficult to actively replicate and occur during functional movement. Abnormal "end feels" of movement have been found to be associated with a patient's concordant sign.[64,65] The clinician should strive to correlate any detection of abnormal movement to the concordant signs.

Passive Physiological Knee Flexion

- Step One: The patient assumes the supine position.
- Step Two: The clinician grasps the patient's leg just proximal to the knee with one hand and just distal to the mid-tibia with the other. The clinician may find it necessary to use a portion of his or her chest to support the leg while applying movement. Additionally, the clinician should strive to maintain the lower extremity/hip in a consistent neutral position (this can be modified as needed to correlate to the concordant signs).
- Step Three: The clinician gently moves the knee into the flexion (sagittal) plane, stopping at the first point of pain.
- Step Four: The clinician will assess for symmetry of motion and assess for reproduction of concordant signs while moving the knee beyond the first point of pain, progressing toward end range as the patient's complaint of pain allows.
- Step Five: Repeated movements or sustained holds into knee flexion are then performed to the point of flexion range that pain allows to assess the response of the repeated movements on the patient's pain (or like symptoms).
- Step Six: As needed, the clinician can apply force (overpressure) to the end of range to assess for pain response.

Figure 14.16 Passive Physiological Knee Flexion

Knee Flexion with Abduction and Adduction

Combined passive movements are used to examine further available physiological range in an effort to find concordant signs and improve the sensitivity of the examination.

- Step One: The clinician follows the procedures above and flexes the knee to 10–20 degrees short of end of available range.
- Step Two: The clinician firmly grasps the distal femur with one hand while holding the lateral femur against his or her chest. This handling will help ensure that the clinician *does not allow femoral rotation* during this phase of the examination. The clinician's other hand will grasp the distal tibia.
- Step Three: As the clinician flexes the knee, he or she directs the heel toward the direction of the greater trochanter of the hip (producing abduction movement of the tibia). Movement will occur up until pain is reproduced, then gently past this point as tolerated. If the concordant sign is not reproduced, this movement can then be repeated with the tibia held in various degrees of available tibia internal rotation.

Figure 14.17 Knee Flexion and Abduction

- Step Four: As needed, the clinician can apply force (overpressure) to the end of range to assess for pain response.
- Step Five: As the clinician flexes the knee, he or she directs the heel toward the direction of the groin (producing adduction movement of the tibia). Movement will occur up until pain is reproduced, then gently past this point as tolerated. If the concordant sign is not reproduced, this movement can then be repeated with the tibia held in various degrees of available tibia external rotation.
- Step Six: As needed, the clinician can apply force (overpressure) to the end of range to assess for pain response.

Figure 14.18 Knee Flexion with Adduction

Passive Physiological Extension

Range of extension ROM is often much smaller than flexion, thus handling procedures are very different.

- Step One: The patient assumes the supine position.
- Step Two: The clinician grasps the lateral ankle with one hand, while the interthenar eminence of the other hand is placed on the tibial tubercle.
- Step Three: Using side bending of the trunk and a simultaneous force on the above-noted contact areas with the hands, the clinician then produces an extension movement of the knee, stopping at the first point of pain.
- Step Four: The clinician will assess for symmetry of motion and assess for reproduction of concordant signs while moving the knee beyond the first point of pain, progressing toward end range as the patient's complaint of concordant pain allows.
- Step Five: Repeated movements into knee extension are then performed to the point of extension range that pain allows to assess the response of the repeated movements on the patient's pain (or like symptoms).
- Step Six: As needed, the clinician can apply force (overpressure) to the end of range to assess for pain response.

Figure 14.19 Passive Knee Extension

Knee Extension with Adduction and Abduction

As noted previously, combined passive movements are used to examine further available physiological range in an effort to find concordant signs and improve the sensitivity of the examination. The following example is for knee extension combined movements:

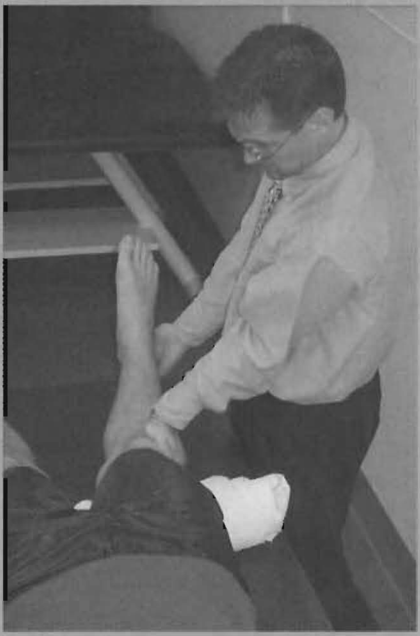

Figure 14.20 Knee Extension with Abduction

- Step One: The clinician follows the procedures above and extends the knee to 10–15 degrees short of end of available range.
- Step Two: The clinician grasps the lateral ankle as noted above, but will move the interthenar eminence just lateral to the tibial tubercle.

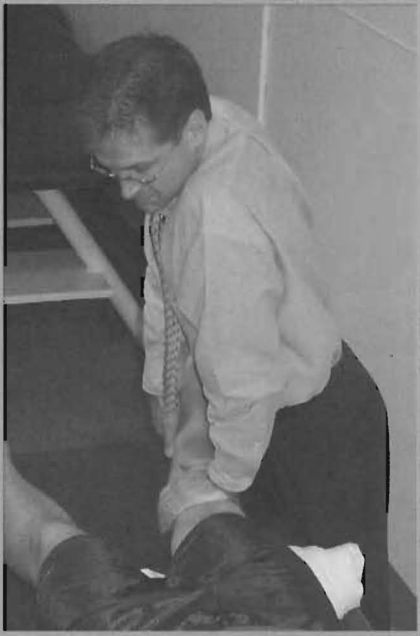

Figure 14.21 Knee Extension with Adduction

- Step Three: As the clinician extends the knee, using side bending of the trunk and a simultaneous force on these contact areas with the hands, the clinician then produces an extension–abduction movement of the knee (abduction movement of the tibia). Movement will occur up until pain is reproduced, then gently past this point as tolerated.
- Step Four: As needed, the clinician can apply force (overpressure) to the end of range to assess for pain response.
- Step Five: The clinician grasps the lateral ankle as noted above, but will move the interthenar eminence just medial to the tibial tubercle.
- Step Six: As the clinician extends the knee, using side bending of the trunk and a simultaneous force on these contact areas with the hands, the clinician then produces an extension–adduction movement of the knee (adduction movement of the tibia). Movement will occur up until pain is reproduced, then gently past this point as tolerated.
- Step Seven: As needed, the clinician can apply force (overpressure) to the end of range to assess for pain response.

Tibial Internal and External Rotation

Approximately 20 degrees of rotation occurs at the tibial femoral joint. Restrictions are common and may require physiological intervention.

- Step One: The patient assumes a supine position. The clinician cradles the tibia on his or her forearm and grasps the foot distally at the heel.
- Step Two: The ankle of the patient is passively dorsiflexed to create a stable lever during rotation. The knee is flexed to approximately 90 degrees.
- Step Three: The clinician applies a passive internal rotation to the first point of pain. Repeated movements or sustained holds are applied to determine if the movements reduce pain. The activity is repeated toward end range.

Figure 14.22 Tibial Internal Rotation at 90 Degrees of Knee Flexion

- Step Four: The clinician applies a passive external rotation to the first point of pain. Repeated movements or sustained holds are applied to determine if the movements reduce pain. The activity is repeated toward end range.

Figure 14.23 Tibial External Rotation at 90 Degrees of Knee Extension

Summary

- The purpose of passive physiological testing is to reproduce the concordant sign of the patient.
- By incorporating adduction, abduction, and rotations in the plane-based movements of flexion and extension, the clinician may more accurately isolate the disorder of the patient.

Passive Accessory Movements

Assessment of passive accessory movements is used to assess component motion of a joint segment/region, including the relevance to the patient's impairment and concordant signs. As we have repeated throughout this text, assessment of position, orientation, or movement of the body/joint segments without inclusion of the relevance to the concordant signs will yield less reliable clinical information and will limit the clinician's ability to analyze further the patient's response to movements.

The following describes the procedures of passive accessory movement testing of the knee. It is important to note that the assessment of accessory glides necessitates movements in varying ranges of physiological ROM. Careful analysis will allow the clinician to establish the relevance of an impairment of accessory motion to physiological ROM and to the concordant signs that may have been established in the active and/or passive physiological components of the examination.

Tibiofemoral Joint—Posterior to Anterior

- Step One: The patient assumes the supine position. The knee is prepositioned at 60–80 degrees of flexion.
- Step Two: The clinician will grasp the proximal tibia with both hands, wrapping the fingers around to the posterior tibia.
- Step Three: The clinician then gently moves the tibia on the femur in a posterior-to-anterior direction, stopping at the first point of pain. The pain is assessed for the concordant sign. In knee flexion ranges of less than approximately 60 degrees, the clinician will need to ensure he or she moves his or her body from posterior to anterior and slightly caudal to attempt not to passively flex the patient's knee. Additionally, the clinician will need to ensure to maintain a consistent position of the hip for all ranges assessed.
- Step Four: The clinician will assess for quality of motion and assess for reproduction concordant signs while moving the knee beyond the first point of pain, progressing toward end range as the patient's complaint of pain allows.
- Step Five: Repeated movements are then performed to the point of range/movement that pain allows, to assess the response of the repeated movements on the patient's pain (or like symptoms).

Figure 14.24 Posterior-Anterior Mobilization of the Tibiofemoral Joint

Anterior-to-Posterior Mobilization of the Tibiofemoral Joint

- Step One: The patient assumes the supine position. The clinician can place a small bolster behind the femur to allow the knee to rest in approximately 10–30 degrees of flexion.
- Step Two: The clinician will place both thumbs on the tibial tubercle and allow the fingers to rest on the postero-lateral aspects of the tibia.
- Step Three: Pushing with the thumbs, the clinician then gently moves the tibia on the femur in an anterior-to-posterior direction, stopping at the first point of pain.
- Step Four: The clinician will assess for quality of motion and assess for reproduction of concordant signs while moving the knee beyond the first point of pain, progressing toward end range as the patient's complaint of pain allows.
- Step Five: Repeated movements are then performed to the point of range/movement that pain allows, to assess the response of the repeated movements on the patient's pain (or like symptoms).

Figure 14.25 Anterior to Posterior Mobilization of the Tibiofemoral Joint

Medial and Lateral Shear of the Tibiofemoral Joint

- Step One: The patient assumes the supine position. The clinician can place a small bolster behind the femur to allow the knee to rest in approximately 10–20 degrees of flexion.
- Step Two: For medial shear, the clinician uses one hand to grasp the medial aspect of the distal femur, in the region of the condyles, while with the other hand the clinician grasps the lateral aspect of the proximal tibia.
- Step Three: The clinician will stabilize the femur while applying a medially directed movement of the tibia on the femur, stopping at the first point of pain.

Figure 14.28 Medial Glide of the Tibia on the Femur

(*continued*)

- Step Four: The clinician will assess for quality of motion and assess for reproduction of concordant signs while moving the knee beyond the first point of pain, progressing toward end range as the patient's complaint of pain allows.
- Step Five: Repeated movements are then performed to the point of range/movement that pain allows, to assess the response of the repeated movements on the patient's pain (or like symptoms).
- Step Six: For lateral shear, the clinician uses one hand to grasp the lateral aspect of the distal femur, in the region of the condyles, while with the other hand the clinician grasps the lateral aspect of the proximal tibia.
- Step Seven: The clinician will stabilize the femur while applying a laterally directed movement of the tibia on the femur, stopping at the first point of pain.

Figure 14.29 Lateral Glide of the Tibia on the Femur

The movement of the tibia on the femur may isolate the outer horns of the menisci and force movement of the menisci. Movement of the tibia posteriorly on the femur results in anterior movement of the menisci on the femur. Figure 14.26 illustrates this process.

The movement of the tibia anteriorly on the femur in a flexed position may force posterior migration of the meniscus with respect to the tibia. Figure 14.27 illustrates this phenomenon.

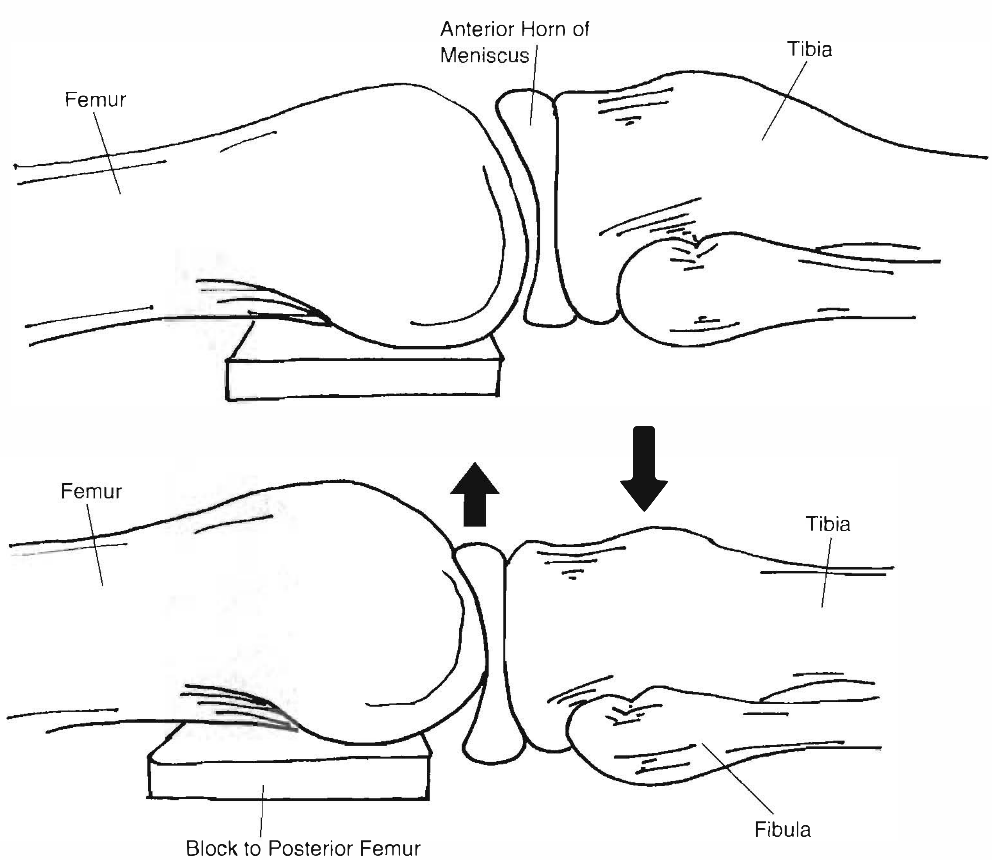

Figure 14.26 Theoretical Movement of the Anterior Horn of the Meniscus

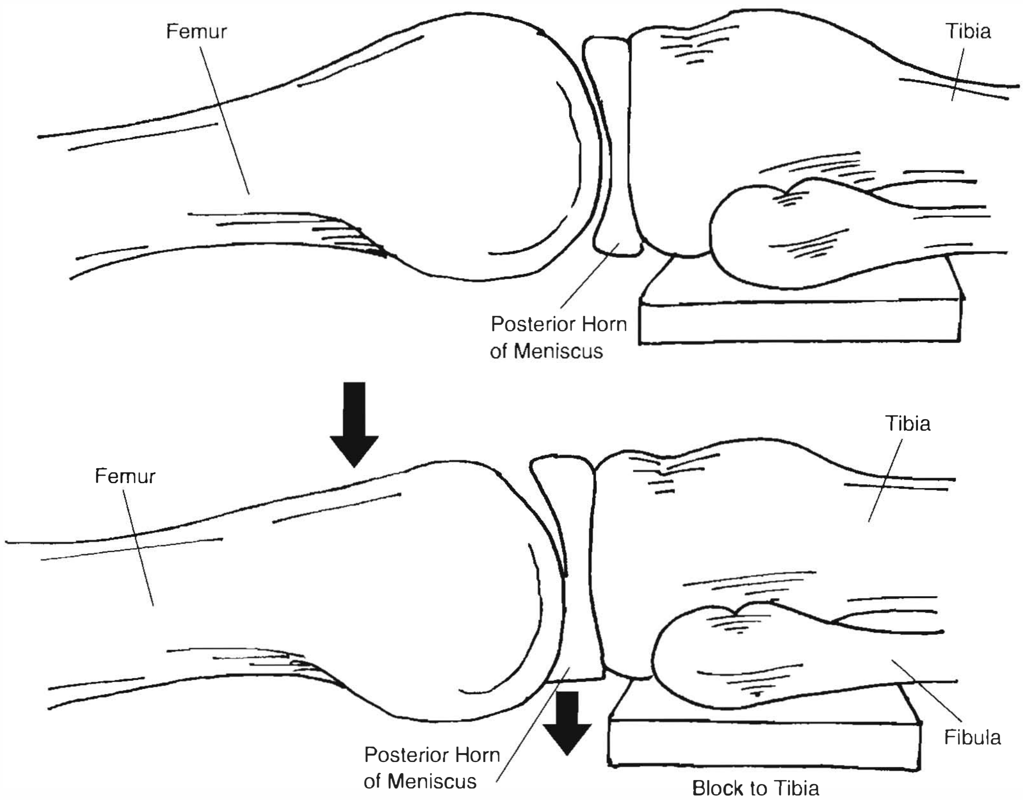

Figure 14.27 Theoretical Movement of the Posterior Horn of the Meniscus

Rotations of the Tibiofemoral Joint

- Step One: The patient assumes a hooklying position. The clinician stabilizes the foot by sitting on the dorsum.
- Step Two: The clinician grabs the lateral half of the tibia with one hand and stabilizes the femur with the other. The clinician then applies a medially directed and anterior rotation to the lateral half of the tibia.

Figure 14.30 Medial and Anterior Rotation of the Tibia on the Femur

(*continued*)

- Step Three: The clinician grabs the medial half of the tibia with one hand and stabilizes the femur with the other. The clinician then applies a lateral and anterior rotation to the lateral half of the tibia.

Figure 14.31 Lateral and Anterior Rotation of the Tibia on the Femur

- Step Four: The clinician grabs the lateral half of the tibia with one hand and stabilizes the femur with the other. The clinician then applies a medial and posterior rotation to the lateral half of the tibia.

Figure 14.32 Medial and Posterior Rotation of the Tibia on the Femur

- Step Five: The clinician grabs the medial half of the tibia with one hand and stabilizes the femur with the other. The clinician then applies a lateral and posterior rotation to the lateral half of the tibia.
- Step Six: Movements are repeated or sustained (at the first point of pain and at end range) to determine if concordant and whether the pain abolishes with movements.

Figure 14.33 Lateral and Posterior Rotation of the Tibia on the Femur

During movement from flexion to extension, the patella starts in a medially tilted position and then shifts to neutral, lastly shifting to a laterally tilted position while in extension.[48] Additionally, the patella moves distally during progressive extension to flexion, variably moves from anterior to posterior movement (in many cases the patella just moved posteriorly), and variably moves progressively laterally during knee flexion.[3] Consequently, it is important to assess the numerous ranges of patella mobility.

Cephalad and Caudal Movements of the Patellofemoral Joint

- Step One: The patient assumes the supine position. The knee is placed in approximately 30 degrees of flexion.
- Step Two: The clinician places the apex of the patella in their interthenar groove, cupping the hand slightly. Additionally, the clinician will lean forward and align his or her forearm with the shaft of the tibia.
- Step Three: With his or her other hand, the clinician places the web space of his or her hand across the superior aspect of the patella. Additionally, the clinician will lean forward and align his or her forearm with the shaft of the femur.
- Step Four: The clinician then gently glides the patella in cephalad direction, stopping at the first point of pain.
- Step Five: The clinician will assess for quality of motion and assess for reproduction of concordant signs while moving the patella beyond the first point of pain, progressing toward end range as the patient's complaint of pain allows.
- Step Six: Repeated movements are then performed to the point of range/movement that pain allows, to assess the response of the repeated movements on the patient's pain (or like symptoms).

Figure 14.34 Cephalic Glide of the Patella

- Step Seven: *Caudal Movement.* The clinician can follow the steps for cephalad movement of the patella, but will move the patella in a caudal direction.

Figure 14.35 Caudal Glide of the Patella

Medial and Lateral Movements of the Patellofemoral Joint

- Step One: The patient assumes the supine position. The knee is placed in approximately 30 degrees of flexion.
- Step Two: *Lateral Movement.* The clinician will stand on the lateral side of the knee and place the fingers against the medial border of the patella.
- Step Three: The clinician will produce a laterally directed movement of the patella, stopping at the first point of pain.
- Step Four: The clinician will assess for quality of motion and assess for reproduction concordant signs while moving the patella beyond the first point of pain, progressing toward end range as the patient's complaint of pain allows.
- Step Five: Repeated movements are then performed to the point of range/movement that pain allows, to assess the response of the repeated movements on the patient's pain (or like symptoms).

Figure 14.36 Lateral Glide of the Patella

- Step Six: *Medial Movement.* The clinician can follow the above steps for lateral movement of the patella (although the thumbs and web space now provide the contact force), but produce movement in a medially directed direction.

Figure 14.37 Medial Glide of the Patella

Medial and Lateral Tilt Movements of the Patellofemoral Joint

Based on the posterior facet anatomy of the patella, medial and lateral "tilting" or rotation of the patella in the transverse plane is an accessory movement of the patellofemoral joint.

- Step One: The patient assumes the supine position. The knee is placed in approximately 30 degrees of flexion.
- Step Two: To create a medial tilt, the clinician will place one of his or her thumbs or thenar eminences on the medial half of the patella, while the other thumb or thenar eminence blocks the lateral aspect of the patella (to limit the patella from moving laterally).
- Step Three: The clinician will produce a medial "tilting" movement of the patella by pushing on the medial half of the patella in an anterior-to-posterior direction, stopping at the first point of pain.
- Step Four: The clinician will assess for quality of motion and assess for reproduction of concordant signs while moving the patella beyond the first point of pain, progressing toward end range as the patient's complaint of pain allows.
- Step Five: Repeated movements are then performed to the point of range/movement that pain allows, to assess the response of the repeated movements on the patient's pain (or like symptoms).

Figure 14.38 Medial Tilt of the Patella

- Step Six: To create a lateral tilt, the clinician can follow the above steps, but will produce lateral tilting movement by producing movement in an anterior-to-posterior direction on the lateral half of the patella. The clinician should also be sure to block the medial border of the patella to prevent a medial movement of the patella during the lateral tilting movement.

Figure 14.39 Lateral Tilt of the Patella

Medial and Lateral Rotation Movements

The medial and lateral rotation described in this section refers to rotation in the frontal plane (i.e., the apex [superior border] of the patella directed medially is medial rotation).

- Step One: The patient assumes the supine position. The knee is placed in approximately 30 degrees of flexion.
- Step Two: *Medial Rotation*. The clinician, standing on the lateral side of the knee, will grasp the patella with both hands, placing the apex of the patella in the web space of the caudal hand and the superior aspect of the patella in the web space of the cephalad hand.
- Step Three: To produce a medially directed rotation, the clinician will rotate the apex of the patella medially, stopping at the first point of pain. The clinician may find it helpful when producing these movements to take up tissue slack around the knee. This can be accomplished by placing the hands on the patella as described in Step Two (the following describes medially directed movement) but place the cephalic hand medially (about the 2 o'clock position) and the caudal hand just opposite the cephalad hand (about the 8 o'clock position). Now firmly grasping the soft tissues move your hands back to the original position. This technique should allow the clinician to feel more localized movement of the patella versus the surrounding soft tissues.
- Step Four: The clinician will assess for quality of motion and assess for reproduction of concordant signs while moving the patella beyond the first point of pain, progressing toward end range as the patient's complaint of pain allows.

Figure 14.40 Medial Rotation of the Patella

- Step Five: Repeated movements are then performed to the point of range/movement that pain allows, to assess the response of the repeated movements on the patient's pain (or like symptoms).
- Step Six: *Lateral Rotation*. The clinician can follow the above steps but produce movement of the patella in a lateral direction.

Figure 14.41 Lateral Rotation of the Patella

Patellar Compression

Compression of joint surfaces in examination is purported to aid in determining if a patient's symptoms are from a "joint surface disorder"; that is, possible changes to the joint surfaces that may contribute to a patient's symptoms.[66] There is no known validity to the fact that pain from just compression of a joint surface correlates to tissue changes of the joint surfaces. With this in mind, the reader is encouraged to consider compression as an additional examination technique that may reveal concordant signs. Additionally, compression can be used to aid in ruling out the involvement of a joint region when examination movements are assessed with strong compression without symptom provocation.[66]

Common clinical findings that suggest examination with compression include:[66]

1. The standard examination movements do not clearly correlate to the patient's symptoms.
2. The patient complaint of pain through range of movement.
3. Crepitus of the joint with movement.
4. Heavy work or activity causes minor symptoms ("compressive loads").
5. Lying on the involved joint provokes symptoms.

There are varieties of joint regions that require examination with compression, the patellofemoral joint being one of them. For the patellofemoral joint, cephalad/caudal, medial/lateral, medial/lateral tilt, and medial/lateral rotation movements are performed with compression. The clinician can perform the previously described examination movements of the patellofemoral joint with compression then assess for relevance to concordant signs. Compression is achieved by moving the patella toward the femur, gently at first, then producing the previously described movements. As with any other accessory examination techniques, the clinician should consider assessing with the knee positioned in various ranges of physiological ROM.

Patellar Distraction

Patellar distraction is basically the opposite of compression. Gently, the clinician will need to use the pads of the fingers to grasp the lateral and medial posterior facet-region of the patella and produce a movement of the patella away from the femur. Assessment then includes producing cephalad/caudal, medial/lateral, medial/lateral tilt, medial/lateral rotation movements while sustaining distraction. The clinician's goals will remain the same of relating the relevance of provoking movements to concordant signs.

Posterior-to-Anterior Glide of Superior Tibial–Fibular Joint

The superior tibial–fibular joint region should be included in a comprehensive evaluation of the knee as this structure can contribute to pain and dysfunction in the knee region and plays an integral role in the interaction between the foot/ankle and the knee.

- Step One: The patient assumes the sidelying position with the area examined facing upward. The clinician should aim to position the lower extremity in a relatively neutral position (i.e., pillow between knees, etc.)
- Step Two: The clinician will stand behind the patient and palpate for the posterior margin of the head of the fibula.
- Step Three: The clinician will then place the pads of his or her thumbs against the posterior margin of the head of the fibula.
- Step Four: Using the thumbs, produce a posterior-to-anterior directed movement, stopping at the first point of pain.
- Step Five: The clinician will assess for quality of motion and assess for reproduction of concordant signs while moving the fibula beyond the first point of pain, progressing toward end range as the patient's complaint of pain allows.
- Step Six: Repeated movements are then performed to the point of range/movement that pain allows, to assess the response of the repeated movements on the patient's pain (or like symptoms).
- Step Seven: The above-described movement should then be performed with compression. This is achieved by leaving one thumb on the posterior margin of the fibula and taking the interthenar groove of the other hand and placing it on top of the thumb. Compression is achieved by pushing downward toward the table while simultaneously producing the posterior-to-anterior movement. Steps Five and Six are then followed while using compression.

Figure 14.42 Posterior to Anterior Glide of the Tibial-Fibular Joint

Anterior-to-Posterior Glide of the Tibial–Fibular Joint

- Step One: The patient assumes the sidelying position with the area examined facing upward. The clinician should aim to position the lower extremity in a relatively neutral position (i.e., pillow between knees, etc.).
- Step Two: The clinician stands in front of the patient and palpates for the posterior margin of the head of the fibula.
- Step Three: The clinician then places the pads of his or her thumbs against the anterior margin of the head of the fibula.
- Step Four: Using the thumbs, produce a posterior-to-anterior directed movement, stopping at the first point of pain.
- Step Five: The clinician will assess for quality of motion and assess for reproduction of concordant signs while moving the fibula beyond the first point of pain, progressing toward end range as the patient's complaint of pain allows.
- Step Six: Repeated movements are then performed to the point of range/movement that pain allows, to assess the response of the repeated movements on the patient's pain (or like symptoms).
- Step Seven: The previously described movements should then be performed with compression. This is achieved by leaving one thumb on the anterior margin of the fibula and taking the interthenar groove of the other hand and placing it on top of the thumb. Compression is achieved by pushing downward toward the table while simultaneously producing the anterior-to-posterior movement. Steps Five and Six are then followed while using compression.

Figure 14.43 Anterior to Posterior Glide of the Tibial Fibular Joint

Cephalad and Caudal of the Tibial–Fibular Joint

- Step One: The patient assumes the sidelying position with the area examined facing upward. The clinician should aim to position the lower extremity in a relatively neutral position (i.e., pillow between knees, etc.)
- Step Two: The clinician will stand behind the patient and palpate for the anterior and posterior margins of the head of the fibula.
- Step Three: The clinician then grasps the rear-foot of the same lower extremity. The clinician will ensure that the rear-foot is clear of the table to produce movement of the rear-foot.

Figure 14.44 Cephalic Glide of the Fibula

- Step Four: The clinician will indirectly produce cephalad and caudal movements of the fibula by producing inversion (caudal fibula) or eversion (cephalad fibula) of the rear-foot. Simultaneous palpation of the superior fibula occurs during these movements. Movement occurs up to the first point of pain.
- Step Five: The clinician will assess for quality of motion and assess for reproduction of concordant signs while moving the fibula beyond the first point of pain (by moving the rear-foot), progressing toward end range as the patient's complaint of pain allows.
- Step Six: Repeated movements are then performed to the point of range/movement that pain allows, to assess the response of the repeated movements on the patient's pain (or like symptoms).

Figure 14.45 Caudal Glide of the Fibula

Summary

- The purpose of passive accessory testing is to reproduce the concordant sign of the patient.
- Appropriate isolation of passive accessory testing may require prepositioning of the knee to implicate symptoms.
- Strategies such as compression are effective in isolating painful conditions.
- Combined accessory movements may tighten structures and further implicate casual problems.

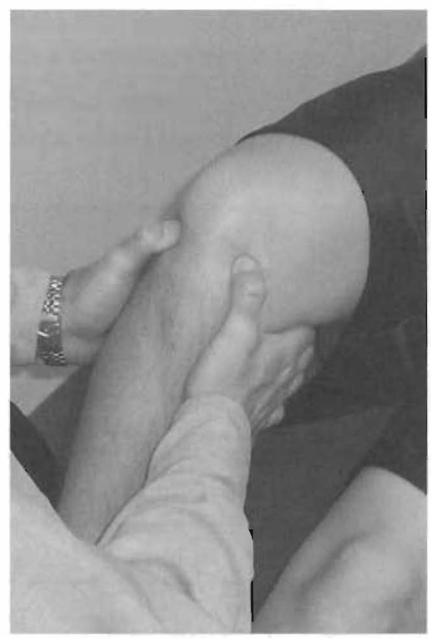

Figure 14.46 Palpation of the Medial Tibial Joint Line

Clinical Special Tests

Palpation

Palpation tests for the knee are best divided into two main categories: (1) palpation for the presence of **intraarticular disorders** and (2) palpation for the presence of a fracture. Table 14.1 outlines the diagnostic value of palpation.

Joint Line Tenderness

Palpation for joint line tenderness (Figure 14.46) is a basic maneuver used to implicate meniscal injuries or general effusion of the knee.[79] Since the anterior–medial aspect of the medial meniscus becomes prominent during internal rotation and flexion, it may be advantageous to preposition the knee during the palpation procedure. Extension of the knee may further improve the sensitivity of this palpation procedure.

Joint Effusion

The benefit of palpation for joint effusion highly depends on the timing and the amount of effusion noted. Calmbach and Hutchens[67] suggest that a "rapid onset" (within 2 hours) of a large, tense effusion suggests rupture of the anterior cruciate ligament or fracture of the tibial plateau with resultant hemarthrosis, whereas a slower onset (24–36 hours) of a mild-to-moderate effusion is consistent with meniscal injury or ligamentous sprain.

Palpation Tests for Knee Fractures

The **Ottawa Knee Rules** are a prospective set of clinical findings that are designed to assist in determining the use of a radiograph. There are five components to the Ottawa Knee Rules: (1) age > 55, (2) tenderness at the head of the fibula, (3) isolated tenderness of patella during

TABLE 14.1 Clinical Special Tests for Palpation

Clinical Test	Author	Sensitivity	Specificity	+LR	−LR
Joint Line Tenderness	Barry et al.[103]	86	43	1.5	0.32
	Noble & Erat[104]	73	13	0.8	2.1
	Fowler & Lubliner[105]	85	30	1.2	0.5
	Saengnipanthkul et al.[106]	58	74	2.2	0.6
	Kurosaka et al.[107]	55	67	1.6	0.67
	Anderson & Lipscomb[108]	77	NR	NA	NA
	Akseki et al. Medial Meniscus[109]	88	44	1.6	0.27
	Akseki et al. Lateral Meniscus[109]	67	80	3.4	0.41
	Karachalios et al. Medial Meniscus[110]	71	87	5.5	0.33
	Karachalios et al. Lateral Meniscus[110]	78	90	7.8	0.24
	Eren Medial Meniscus[111]	86	67	2.6	0.20
	Eren Lateral Meniscus[111]	92	97	30.7	0.08
Joint Effusion	Noble & Erat[104]	53	54	1.1	0.87
	Barry et al.[103]	30	100	NA	NA

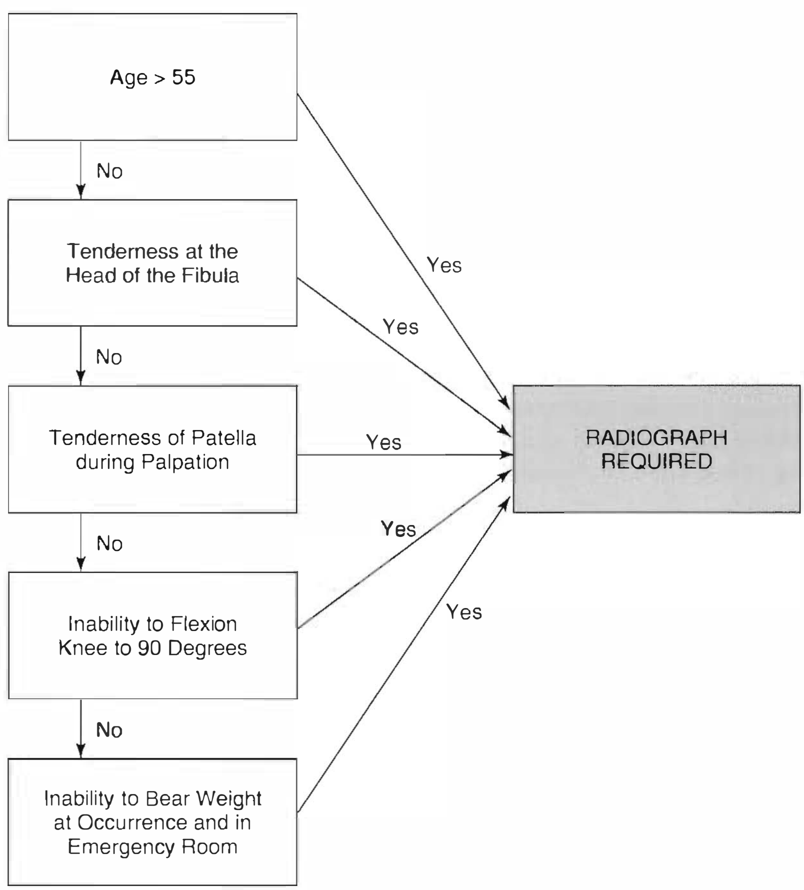

Figure 14.47 The Ottawa Knee Rules Diagram[114]

TABLE 14.2 Clinical Special Tests for Detection of Knee Fractures

Clinical Test	Author	Sensitivity	Specificity	+LR	−LR
Ottawa Ankle Rules	Stiell et al.[112]	100	49.5	NA	NA
	Richman et al.[113]	84.6	45.2	1.54	0.34
	Emparanza & Aginage[114]	100	52	NA	NA
	Szucs et al.[115]	100	46.6	NA	NA
	Stiell et al.[116]	100	48	NA	NA
	Ketelslegers et al.[117]	100	31.6	NA	NA
	Seaberg et al.[118]	96.6	26.5	1.31	0.12
	Tigges et al.[119]	97.7	19.1	1.71	0.12
	Khine et al.[120]	92.3	48.9	1.80	0.15

palpation, (4) inability to flex knee to 90 degrees, and (5) inability to bear weight both immediately and in the emergency department. The Ottawa Knee Rules are recognized as the most valid mechanism to determine whether plain film radiographs are necessary in the event of trauma (Figure 14.47 and Table 14.2).[1]

Manual Muscle Testing

Manual [illegible] with a hand-held dynamometer has demonstrated good reliability in studies: (1) that have used children as subjects,[68] (2) subjects status post hip fracture,[69] (3) patients with cerebral palsy,[70] (4) patients with spinal muscular atrophy,[71] and (5) the community-dwelling elderly.[72] However, manual muscle testing was considered poor when tested using subjects that were healthy.[73] The inclined squat strength test is considered an alternative to the break test and is considered more functional. The test has demonstrated good reliability using subjects with no known pathology.[74]

Clinical Special Tests for Intraarticular Disorders

McMurray's Test

McMurray's test for meniscal tears was first described by McMurray in 1942.[75] Since the development of the test, numerous iterations have been demonstrated, all with the name "McMurray's test".[76] Nonetheless, the true McMurray's test has rarely been studied for diagnostic accuracy and instead several variations are reported as homogeneous adaptations. According to McMurray, external tibial rotation tests the internal (medial) cartilage and internal rotation tests the external (lateral) cartilage. Variations of knee flexion purportedly alter the effect on the anterior and posterior cartilage horns. In theory, greater flexion affects the posterior horn while less knee flexion targets the anterior horn. A positive test is the concordant pain and a click during rotation of the tibia on the femur. Table 14.3 (page 527) outlines the diagnostic value of the test.

- Step One: The patient lies in a supine position. The knee is fully flexed to end range.
- Step Two: The clinician grasps the foot of the patient at the heel to guide tibial rotation. The other hand of the clinician is placed on the top of the patient's knee to counter the force through the heel.
- Step Three: While maintaining knee flexion, the clinician applies medial and lateral rotation of the tibia on the femur. This position purportedly tests the posterior horns of the medial and lateral meniscus.
- Step Four: The clinician may vary the angle of knee flexion to target different regions of the meniscus. The more the knee is placed in extension the greater the chance of targeting the anterior horn of the meniscus (purportedly).
- Step Five: A positive test includes concordant pain and a clicking sound either heard or felt.

Figure 14.48 McMurray's Test

Apley's Compression Test

Apley first described Apley's compression, also known as Apley's grind test, in 1947.[77] The test originally described by Apley involves several motions that involve both compression and distraction forces. The Apley's compression test described by Hoppenfield[78] involves compression and twisting forces performed at the knee in absence of the flexion and extension movements. Distraction forces described by Apley are followed to differentiate between meniscal and capsuloligamentous injuries. For simplicity, the compression test as described by Hoppenfield is reported.[78] Table 14.3 outlines the diagnostic value of the test.

- Step One: The patient lies in a prone position. The knee is flexed to approximately 90 degrees.
- Step Two: The clinician grasps the foot of the patient and applies a compressive force through the shaft of the tibia.
- Step Three: While maintaining the compressive force, the clinician applies a medial and lateral twist of the tibia on the stabilized femur.
- Step Four: The clinician determines if the compression force is concordant and painful.
- Step Five: The clinician then places his or her knee on the patient's thigh to support it to the table. A distraction force is then applied.
- Step Six: During the maintained distraction force, the clinician applies medial and lateral twisting to the knee to determine if the action is painful and concordant.
- Step Seven: According to Apley, if the knee is painful and concordant during compression and twisting, the problem is associated with the meniscus. If the knee is painful and concordant during the distraction and twisting, the problem is capsuloligamentous.

Figure 14.49 Apley's Compression Test

Figure 14.50 Apley's Distraction Test

TABLE 14.3 Clinical Special Tests for Detection of Intraarticular Disorders

Clinical Test	Author	Sensitivity	Specificity	+LR	−LR
McMurray's Test	Kurosaka et al.[107]	37	77	1.61	0.81
	Noble et al.[104]	63	57	1.5	0.65
	Fowler & Lubliner[105]	29	96	7.8	0.73
	Evans et al.[121]	20	91	2.3	0.87
	Saengnipanthkul et al.[106]	47	94	7.8	0.56
	Anderson and Lipscomb[108]	58	NR	NA	NA
	Akseki et al. Medial Meniscus[109]	67	69	2.16	0.47
	Akseki et al. Lateral Meniscus[109]	53	88	4.41	0.53
	Karachalios et al. Medial Meniscus[110]	48	94	8	0.55
	Karachalios et al. Lateral Meniscus[110]	65	86	4.64	0.40
Apley's Compression Test	Fowler & Lubliner[105]	16	80	0.8	1.1
	Grifka et al.[122]	58	80	2.9	0.5
	Kurosaka et al.[107]	13	90	1.3	0.96

Clinical Tests for Anterior Cruciate Instability

Anterior Drawer Test

Numerous studies have investigated the diagnostic accuracy of the anterior drawer test. The majority have noted usefulness of the test, although the diagnostic figures are not as strong as the Lachman's test. The test was originally described by Segund in 1879.[79] It is notable that the sensitivity of the anterior drawer test is affected by concomitant injuries.[79] A positive test is considered as "abnormal" displacement of the tibia on the femur when compared to the opposite extremity.[79] Table 14.4 (page 529) outlines the diagnostic value of the test.

- Step One: The patient lies in a supine position. The hip is flexed to 45 degrees and the knee is flexed to 90 degrees.
- Step Two: The clinician sits at the foot of the patient, stabilizing the lower extremity by sitting slightly on the foot. The hands of the clinician are placed behind the tibia of the patient. The thumbs are placed superiorly to the joint line, on the femur.
- Step Three: The clinician applies a superior force to the hamstring tendons to inhibit contraction and to feel for relaxation.
- Step Four: The clinician applies quick posterior-to-anterior movements to determine the displacement of the tibia on the fixed femur.
- Step Five: The opposite extremity is evaluated for comparison.

Figure 14.51 The Anterior Drawer Test

Lachman's Test

Torg et al.[80] first described Lachman's test. The test is also a clinical assessment for anterior cruciate ligament disruption. The test demonstrates very good diagnostic values, although some examiners with small hands may have difficulties performing the procedure. Additionally, the procedure is difficult to perform on patients with large thigh girths. A positive test is also considered "abnormal" displacement of the tibia on the femur when compared to the opposite extremity.[79] Table 14.4 outlines the diagnostic value of the test.

- Step One: The patient lies in a supine position. The knee is placed in approximately 15 degrees of knee flexion.
- Step Two: The clinician sits to the outside of the patient's knee. The femur is stabilized caudally with one hand while the other cradles the tibia proximally and posteriorly. For smaller clinicians, it may be helpful to plant the elbow of the tibial-hand on the mat for a counterforce.
- Step Three: The clinician applies quick posterior-to-anterior thrust to determine the displacement of the tibia on the fixed femur.
- Step Four: The opposite extremity is evaluated for comparison.

Figure 14.52 Lachman's Test

Pivot Shift Test

The action of the pivot shift is similar to the movement of the lower extremity during giving way of the knee.[79] The pivot shift movement includes an anterior subluxation of the lateral tibial plateau on the femur when the knee is in full flexion and a reduction of the translation when the knee approaches full extension. A positive test is the clunking and the pain that occurs during the reduction at approximately 30 degrees short of extension. Table 14.4 outlines the diagnostic value of the test.

- Step One: The patient lies in a supine position. The affected leg is picked up by the clinician at the ankle with one hand and the knee is supported with the other at the head of the fibula.
- Step Two: The clinician places the knee in at least 90 degrees of flexion. The knee is guided into extension with both hands while the clinician places a slight valgus force at the fibular head.
- Step Three: The valgus force is increased as the knee approaches full extension.
- Step Four: If positive, the displaced tibia will reduce at approximately 30 degrees of extension.
- Step Five: The opposite extremity is evaluated for comparison.

Figure 14.53 The Pivot Shift Test

TABLE 14.4 Clinical Special Tests for Detection of Anterior Cruciate Ligament Instability

Clinical Test	Author	Sensitivity	Specificity	+LR	−LR
Anterior Drawer Test	Hardaker et al.[123]	18	NR	NA	NA
	Tonino et al.[124]	27	98	13.5	0.74
	Boeree & Ackroyd[125]	56	92	7	0.47
	Richter et al.[126]	67	88	5.6	0.37
	Steinbruck & Wiehmann[127]	92	91	10.2	0.08
	Sandberg et al.[128]	39	78	1.8	0.78
	Braunstein[129]	91	89	8.27	0.10
	DeHaven[130]	9	NR	NA	NA
	Donaldson et al.[131]	70	NR	NA	NA
	Hughston et al.[81]	65	23	0.84	1.52
	Jonsson et al.[132]	93	NR	NA	NA
	Lee et al.[133]	78	100	NA	NA
	Lui et al.[134]	63	NR	NA	NA
	Mitsou & Vallianatos[135]	72	NR	NA	NA
Lachman's Test	Gurtler et al.[136]	100	NR	NA	NA
	Donaldson et al.[131]	99	NR	NA	NA
	Hardaker et al.[123]	74	NR	NA	NA
	Tonino et al.[124]	89	98	44.5	0.11
	Boeree & Ackroyd[125]	63	90	6.3	0.41
	Lee et al.[133]	90	99	90	0.10
	Richter et al.[126]	93	88	7.7	0.07
	Steinbruck & Wiehmann[127]	86	95	17.2	0.14
	Lui et al.[134]	89	100	NA	NA
Pivot Shift Test	Hardaker et al.[123]	29	NR	NA	NA
	Tonino et al.[124]	18	98	9	0.83
	Boeree & Ackroyd[125]	31	97	10.3	0.71
	Richter et al.[126]	48	97	16	0.53
	Steinbruck & Wiehmann[127]	22	99	22	.78
	Dahlstedt & Dalen[137]	37	NR	NA	NA
	Dehaven[130]	27	NR	NA	NA
	Donaldson et al.[131]	35	NR	NA	NA
	Liu et al.[134]	95	NR	NA	NA

Clinical Tests for Posterior Cruciate Instability

Injuries of the posterior cruciate ligament (PCL) are uncommon but when present lead to significant dysfunction and disability. The PCL restricts posterior translation of the tibia on the femur during weight bearing. Most PCL injuries occur during impact to the tibia and resultant anterior deceleration of the femur on the stationary tibia.

Posterior Drawer Test

The posterior drawer test was originally described by Noulis in 1875.[79] The test is designed to measure posterior displacement of the tibia on a fixed femur and is considered positive if one leg translates posteriorly more so than the other noninvolved side. The test is prone to false positives if the ACL is torn and the tibia normally sits in an anteriorly translated position. Table 14.5 outlines the diagnostic value of the test.

- Step One: The patient lies in a supine position. The hip of the patient is flexed to 45 degrees and the knee of the patient is flexed to 70 degrees.
- Step Two: The clinician sits at the foot of the patient. Both palms lie flush on the anterior surface of the tibia. The fingers wrap posteriorly to stabilize the tibia.
- Step Three: Stabilizing the foot by sitting on it, the clinician then applies a posterior force of the tibia on the fixed femur in quick thrusts.
- Step Four: A positive test is represented by more tibial translation on one side versus another.
- Step Five: The opposite extremity is evaluated for comparison.

Figure 14.54 The Posterior Drawer Test

Posterior Sag Sign

The posterior sag sign was originally described by Robson in 1903.[79] The test is designed to allow visual confirmation of the posterior tibial translation that occurs during a PCL injury. A positive test is significant visual posterior translation during a non-weight-bearing assessment. Table 14.5 outlines the diagnostic value of the test.

- Step One: The patient lies in a supine position. The hip of the patient is flexed to 45 degrees and the knee of the patient is flexed to 70–90 degrees. Both feet are lifted and held in a neutral position from the mat by the clinician.
- Step Two: There are two possible follow-up steps. The clinician may visually assess the tibial translation without force or the clinician may apply a posterior translation of the tibia through the tibia to enhance the sag.
- Step Three: A positive test is when the tibia translates more posteriorly on one side versus another.
- Step Four: Both legs are assessed at the same time.

Figure 14.55 The Posterior Sag Sign

Quadriceps Active Test

The quadriceps active test was recently described by Daniel et al. in 1988.[79] The test measures anterior translation of the tibia during an active quadriceps contraction while in the "drawer position." Anterior translation during the active quadriceps contraction of 2 millimeters or more is considered a positive test. The excellent diagnostic values of Daniel were most likely inflated secondary to the lack of blinding in the study. Table 14.5 outlines the diagnostic value of the test.

- Step One: The patient lies in a supine position. The hip is flexed to 45 degrees and the knee is flexed to 90 degrees.
- Step Two: The clinician sits at the foot of the patient, stabilizing the lower extremity by sitting slightly on the foot.
- Step Three: The patient is directed to actively contract his or her quadriceps. The clinician looks for anterior displacement of the tibia on the femur comparatively to the opposite side.
- Step Four: The clinician applies quick posterior-to-anterior movements to determine the displacement of the tibia on the fixed femur.
- Step Five: The opposite extremity is evaluated for comparison.

Figure 14.56 The Quadriceps Active Test

TABLE 14.5 Clinical Special Tests for Detection of Posterior Cruciate Ligament Instability

Clinical Test	Author	Sensitivity	Specificity	+LR	−LR
Posterior Drawer Test	Baker et al.[138]	86	NR	NA	NA
	Loos et al.[139]	51	NR	NA	NA
	Rubenstein et al.[140]	90	99	90	0.10
Posterior Sag Sign	Rubenstein et al.[140]	79	100	NA	NA
Quadriceps Active Test	Rubenstein et al.[140]	54	97	18	0.47
	Daniel et al.[141]	98	100	NA	NA

Dial Test

The Dial test is designed to determine isolated posterolateral corner instability and to differentiate the laxity from a PCL tear. The Dial test is performed at 30 degrees of knee flexion. If one side exhibits 10–15 degrees more external rotation than the other, the patient most likely exhibits posterolateral corner laxity. However, if this laxity is only present at 30 degrees and not 90 degrees, the injury is concurrent with a PCL rupture.[81]

- Step One: The patient lies in prone. Both knees are flexed to approximately 30 degrees.
- Step Two: Both knees are passively rotated into external rotation. External rotation is compared bilaterally.
- Step Three: Both knees are moved to 90 degrees of flexion and passively externally rotated. The test can be performed in supine as well.

Figure 14.57 The Dial Test Performed in Prone at 90 Degrees

Clinical Tests for Varus and Valgus Laxity

Varus Stress Test

The varus stress test is designed to measure the integrity of structures to stabilize against varus forces. These structures may include the lateral collateral ligament (LCL) and selected internal ligaments such as the PCL and ACL. Typically, the test is performed in both 0 and 30 degrees of flexion. According to Hughston et al.,[82] lateral instability at 0 degrees of flexion is usually associated with a PCL, arcuate ligament, ACL, or some other disruption. The stress test in slight flexion is more associated with an LCL tear. A positive test is considered greater laxity than the opposite extremity. Table 14.6 outlines the diagnostic value of the test.

- Step One: The patient lies in a supine position. The affected leg is flexed to approximately 30 degrees. The clinician then places the foot in his or her axilla for stability.
- Step Two: The clinician places one hand on the lateral aspect of the tibia and the other hand on the medial aspect of the femur. The clinician then preloads the joint into varus.
- Step Three: Further varus force is applied by twisting the body in the direction of more varus.
- Step Four: A positive test is excess gapping into varus.
- Step Five: The test is repeated at 0 degrees of flexion.

Figure 14.58 The Varus Stress Test at 30 Degrees of Knee Flexion

Valgus Stress Test

The valgus stress test is designed to measure the integrity of structures to stabilize against valgus forces. These structures may include the medial collateral ligament (LCL) and similar selected internal ligaments such as the PCL and ACL as the varus stress test. The test is also performed in both 0 and 30 degrees of flexion with findings in the different positions associated with difference structures. According to Hughston et al.,[82] a positive stress test at 30 degrees and a negative test at 0 indicate a MCL tear with a posterior capsule injury. A positive valgus stress test at 0 degrees may suggest a PCL tear and a tear within the posterior capsule but does not usually indicate an ACL tear.[82] A positive test is considered greater laxity than the opposite extremity. Table 14.6 outlines the diagnostic value of the test.

- Step One: The patient lies in a supine position. The affected leg is flexed to approximately 30 degrees. The clinician then places the foot in his or her axilla for stability.
- Step Two: The clinician places one hand on the medial aspect of the tibia and the other hand on the lateral aspect of the femur. The clinician then preloads the joint into varus.
- Step Three: Further varus force is applied by twisting the body in the direction of more valgus. The body motion is opposite of the varus movement.
- Step Four: A positive test is excess gapping into valgus.
- Step Five: The test is repeated at 0 degrees of flexion.

Figure 14.59 The Valgus Stress Test at 0 Degrees of Knee Flexion

TABLE 14.6 Clinical Special Tests for Varus and Valgus Laxity

Clinical Test	Author	Sensitivity	Specificity	+LR	−LR
Valgus Stress Test	Harilainen et al.[142]	86	NR	NA	NA
	Garvin et al.[143]	96	NR	NA	NA
Varus Stress Test	Harilainen et al.[142]	25	NR	NA	NA

Clinical Special Tests for Patellofemoral Disorders

Vastus Medialis Coordination Test

The vastus medialis coordination test was originally described by Souza.[79] A test is considered positive if coordinated knee extension is poor during terminal knee extension. The region assumed responsible is the vastus medialis. Table 14.7 (page 536) outlines the diagnostic value of the test.

- Step One: The patient lies in a supine position. The clinician places his or her fist or thigh under the knee so that the knee is slightly flexed.
- Step Two: The patient is instructed to extend the knee slowly.
- Step Three: The clinician observes for full extension and a full coordinated movement.
- Step Four: A test is considered positive if the knee is not fully extended or if control of the knee was not apparent during the movement.
- Step Five: The opposite extremity is evaluated for comparison.

Figure 14.60 The Vastus Medialis Coordination Test

Patellar Apprehension Test

The patellar apprehension test is designed to determine whether pain and/or apprehension are present during a lateral glide of the patella. Reproduction of pain or apprehension identifies a positive test. Table 14.7 outlines the diagnostic value of the test.

- Step One: The patient lies in a supine position. The clinician places a lateral glide on the knee cap.
- Step Two: The movement is started at 30 degrees. At that range and during concurrent lateral patellar glide, the knee and the hip are flexed together.
- Step Three: A test is considered positive if the knee is not fully extended secondary to fear or if control of the knee was not apparent during the movement.

Figure 14.61 The Patella Apprehension Test

Clarke's Test

Clarke's test for patellofemoral grinding is designed to measure whether the origination of the pain occurs at the articulation of the patella and the femur. Because the test can provide false positives during full extension, the test is best performed in a slight degree of flexion. A positive test is considered concordant reproduction of pain by the patient. Table 14.7 outlines the diagnostic value of the test.

- Step One: The patient lies in a supine position. The affected leg is placed in slight flexion.
- Step Two: The clinician provides an inferior force to the patella. Slight compression is administered.
- Step Three: The patient is instructed to contract his or her quadriceps. A positive test is pain during the quadriceps contraction concurrently during inferior glide and compression of the patella.

Figure 14.62 Clarke's Test for Patella Grinding

Eccentric Step Test

The eccentric step test involves an eccentric load to the knee during a controlled step down. The test is considered positive if pain is reproduced during the activity. Table 14.7 outlines the diagnostic value of the test.

- Step One: The patient stands on a 15 cm step. The patient is instructed to place his or her arms on his or her hips.
- Step Two: The patient is instructed to lower his or her nonaffected leg slowly to the floor. The affected leg eccentrically controls the lowering.
- Step Three: A positive test is concordant reproduction of symptoms.

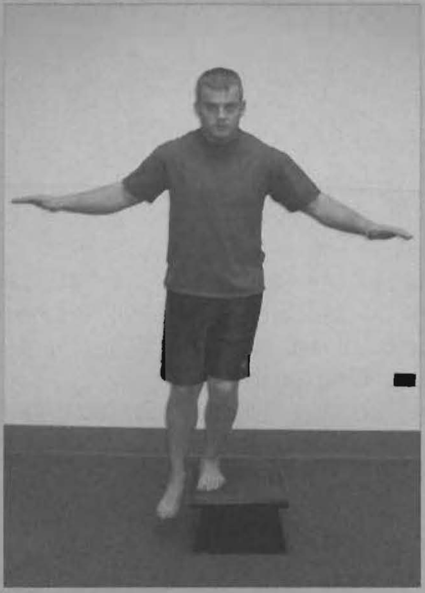

Figure 14.63 The Eccentric Step Test

TABLE 14.7 Clinical Special Tests for Patellofemoral Syndrome

Clinical Test	Author	Sensitivity	Specificity	+LR	−LR
Vastus Medialis Coordination Test	Nils et al.[83]	16.1	92.9	2.26	0.90
Patellar Apprehension Test	Nils et al.[83]	32.3	85.7	2.26	0.78
	Sallay et al.[144]	39	NR	NA	NA
Clarke's Test	Nils et al.[83]	48.4	75	1.94	0.68
Eccentric Step Test	Nils et al.[83]	41.9	82.1	2.34	0.70

Summary

- The purpose of a clinical special test is for confirmation of the examination findings.
- Palpation-based tests may be useful in isolating intraarticular disorders or fractures.
- In general, tests for anterior and posterior laxity demonstrate better likelihood ratios than tests for meniscal involvement.
- Few tests have been studied for the effectiveness of outlining PFPS.

Treatment Techniques

Treatment of knee disorders should focus on identifying physical impairments that are relative to concordant signs. Physical impairments can range from disorders associated with motor control, active and passive mobility, and passive structure competence of the systems of the knee.

Mobilization, active or passive, in any direction may be beneficial for the patient with osteoarthritis.[84] Activities such as active quadriceps contractions have been shown to reduce negative biochemical parameters within the knee.[84] Compression-based mobilizations may be most effective at producing these changes and have been shown to lead to better outcomes than treatments with no compression.[85] As with the previous chapters, the examination methods are often similar to the treatment procedures.

Compression Mobilization of the Tibial–Femoral Joint

- Step One: The patient assumes a prone position. The clinician stabilizes the tibia by applying a perpendicular grip to the tibia and fibula with his or her mobilizing arm.
- Step Two: The knee is bent toward the concordant region of pain. The clinician applies a compressive load through the tibiofemoral joint by loading the heel of the patient with the nonmobilizing hand.
- Step Three: The clinician applies oscillations during compression for treatment of the patient with the mobilizing hand (contact hand of the tibia and fibula).

Figure 14.64 Prone Compression Mobilization of the Tibiofemoral Joint

- Step Four: The procedure can also be performed using similar hand-holds with the patient in a supine position.

Figure 14.65 Supine Compression Mobilization of the Tibiofemoral Joint

Scoop Mobilization for the Posterior Horn

Because the menisci of the knee are frequently injured or fail to move during knee flexion and extension, the menisci are often targeted for treatment. Anterior–posterior mobilizations of the tibia and femur, respectively, should assist in differentiating the anterior and posterior horn of the meniscus. Posterior horn injuries are typically worse during flexion of the knee and functional activities such as squatting. The scoop mobilization is used to target the posterior horn of the meniscus.

- Step One: The patient is placed in a sitting position or a supine, hooklying position.
- Step Two: The clinician flexes the patient to the first point of pain or toward end range flexion.
- Step Three: The clinician places his or her hands at the joint line posteriorly to the knee of the patient. The clinician then further flexes the knee toward end range flexion.
- Step Four: The mobilization procedure is a curvilinear posterior-to-anterior pull that targets the menisci. The technique is most effective near end range flexion.

Figure 14.66 Posterior Anterior Scoop Mobilization

Compression Mobilization of the Patellofemoral Joint

- Step One: The patient assumes a supine position. The knee is flexed slightly by placing a bolster under the knee.
- Step Two: The clinician applies a load to the patella by placing his or her hand on top of the knee cap.
- Step Three: The tibiofemoral joint is oscillated during the load to the knee cap.

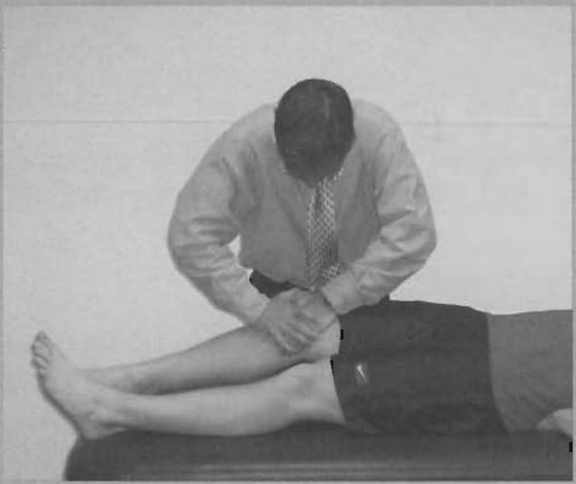

Figure 14.67 Compression Mobilization of the Patellofemoral Joint

Tibial Shear at Multiple Ranges

The tibial shear mobilization is performed at various ranges throughout the knee. Because the capsule is engaged at greater degrees of flexion (Figure 14.68), the tibial shear mobilization in flexion tends to target the capsule. The tibial shear in extension may target the menisci. Concordant pain during passive physiological testing typically identifies the most appropriate position for linear mobilization of the tibia.

Tibial Rotation at Multiple Ranges

Occasionally, rotation is more sensitive at reproducing the concordant sign than the linear shear methods. Subsequently, rotational mobilization may be effective at various ranges as well (Figure 14.69). Like the tibial shear methods, concordant pain during passive physiological testing typically identifies the most appropriate position for rotational mobilization of the tibia.

Figure 14.68 Tibial Shear at 90 Degrees of Flexion

Figure 14.69 Rotation Mobilization at 90 Degrees of Flexion

Hamstring Stretches

Several studies have described various hamstring stretching methods. It appears that any form of hamstring stretching (i.e., hold–relax, static hold) tends to provide the same level of range improvement, with only minimal carryover.[86–88] A hold–relax stretch is an effective method that allows a patient to govern the amount of stretch force provided.[87]

- Step One: The patient assumes a supine position. The hamstrings are prepositioned in a 90/90 posture (hip and knee at 90 degrees).
- Step Two: The clinician places the heel of the patient under his or her shoulder. The patient is instructed to push downward into the shoulder at his or her first point of perceived tightness. Hold times vary, but in most cases, a hold time of 10–15 seconds of a submaxial contraction is appropriate.
- Step Three: The clinician slowly moves the tibia into further extension after each isometric contraction of the patient.

Figure 14.70 Isometric (Hold-Relax) Hamstring Stretching of the Patient

Manipulation of the Tibial Femoral Joint

Meyer et al.[89] describe a tibiofemoral manipulation in which the tibia is rapidly distracted from the femur. The technique was used concurrently with patellofemoral mobilization and leads to a positive consequence in a single case study.

- Step One: The patient assumes a prone position. The clinician uses a belt to stabilize the femur.
- Step Two: The knee is slightly flexed to the targeted range of discomfort. The clinician can preposition the knee in internal or external rotation, depending on the concordant pain of the patient.
- Step Three: The manipulation includes a rapid distraction of the tibia on the femur.

Figure 14.71 Tibiofemoral Distraction Manipulation of the Knee

Summary

- The majority of manual therapy treatment methods are distilled from the examination process.
- Adding components such as compression, end range movements, and distraction may increase the usefulness of manual therapy treatment.

Treatment Outcomes

The literature appears to be lacking in data regarding passive accessory movements of the tibio-femoral joint. Assessment of patellar mobility does appear in the literature as a common component to the treatment of PFPS. Studies have examined the assessment of static patellar orientation, particularly medial and lateral orientation and mobility assessment. With the exception of one study, these assessments demonstrated poor reliability.[50,90–92] One study that did demonstrate acceptable levels of reliability included experienced manual therapists and a single subject.[93]

Evidence exists that rehabilitation, including a variety of methods, is beneficial in the treatment of knee osteoarthritis.[84,94–97] The majority of studies analyzed the effects of various forms of strengthening and stretching exercises. Only a few examined the benefits associated with manual therapy of the knee.[84,94,98]

Noel et al.[85] reported that mobilization with compression, a long axis compression technique of the tibia in a flexed position, led to significantly greater preset range-of-motion goals versus the comparative group who did not receive mobilization with compression. Mobilizations were performed at end range, were concordantly painful, and were rated as unpleasant by more recipients.

Deyle et al.[94] found that treatment that included manual therapy and exercise was more effective than exercise alone. Their findings demonstrated significant improvements in self-perceptions of pain, stiffness, and functional ability as well as the distance walked in 6 minutes when compared to placebo. This implies that these treatment methods have merit and may be beneficial for treatment of patients with osteoarthritis of the knee. It also implies that these treatment methods demonstrate credibility and may lead to faster and more beneficial outcomes. In a follow-up study, Deyle and colleagues[99] reported that an in-clinic program consisting of manual therapy and exercise led to better outcomes than a home-based exercise program in a sample of patients with knee arthritis. The manual therapy included a variety of tibial–femoral and patella mobilizations in conjunction with stretching exercise.

Less evidence exists that manual therapy is beneficial for patients with patellofemoral pain syndrome.[100] Crossley et al.[101] reported improvements in knee flexion during stair climbing compared to placebo treatment. Manual therapy mobilization was a component of treatment but was combined with taping, biofeedback, and strengthening exercises. The *Philadelphia Panel Guidelines* found no evidence to support any form of manual or nonmanual treatment of patellofemoral pain syndrome, specifically friction massage, that demonstrated no benefits in numerous studies.[102]

Summary

- Outside the use of manual therapy for the treatment of arthritis, few studies have examined the effectiveness of treatment for the knee.
- Little evidence exists to support the use of manual therapy for PFPS, although when combined with other treatments, the technique may be beneficial.
- Some evidence exists that include compression with mobilization increases the return of range of motion quicker than an absence of compression.

Chapter Questions

1. How does the biomechanics of the knee factor into the selection of the manual therapy treatment approach?
2. Describe how the femoral condyle dictates the likelihood of PFPS.
3. Outline the method in which a mobilization to the tibia or femur can isolate the menisci.
4. Describe how compression may positively affect a condition during manual therapy of the knee.
5. Describe three variations of a plane-based mobilization that may alter the outcome of the technique.

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